

BFS Traversal

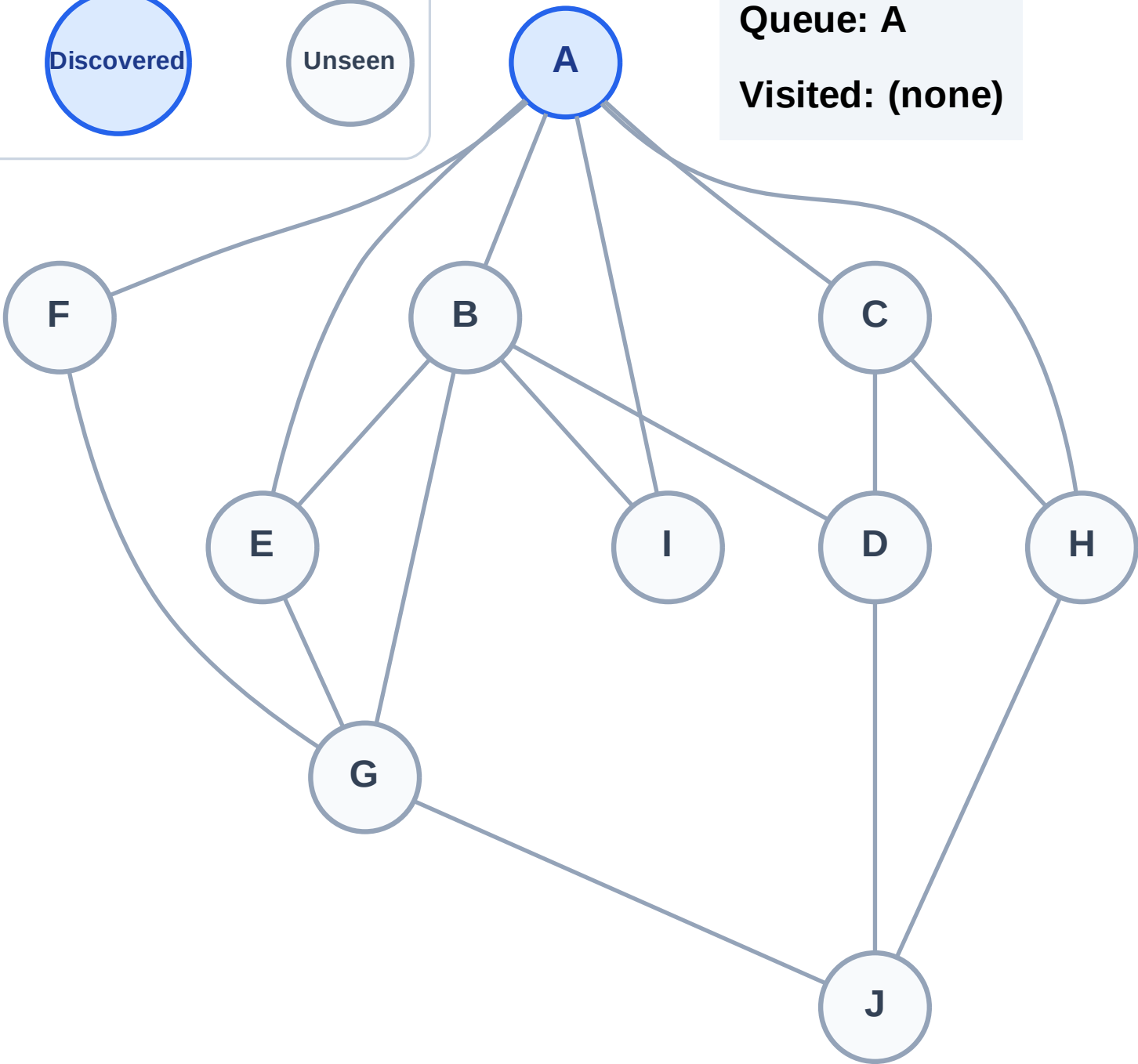
Step 1: Start at A

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

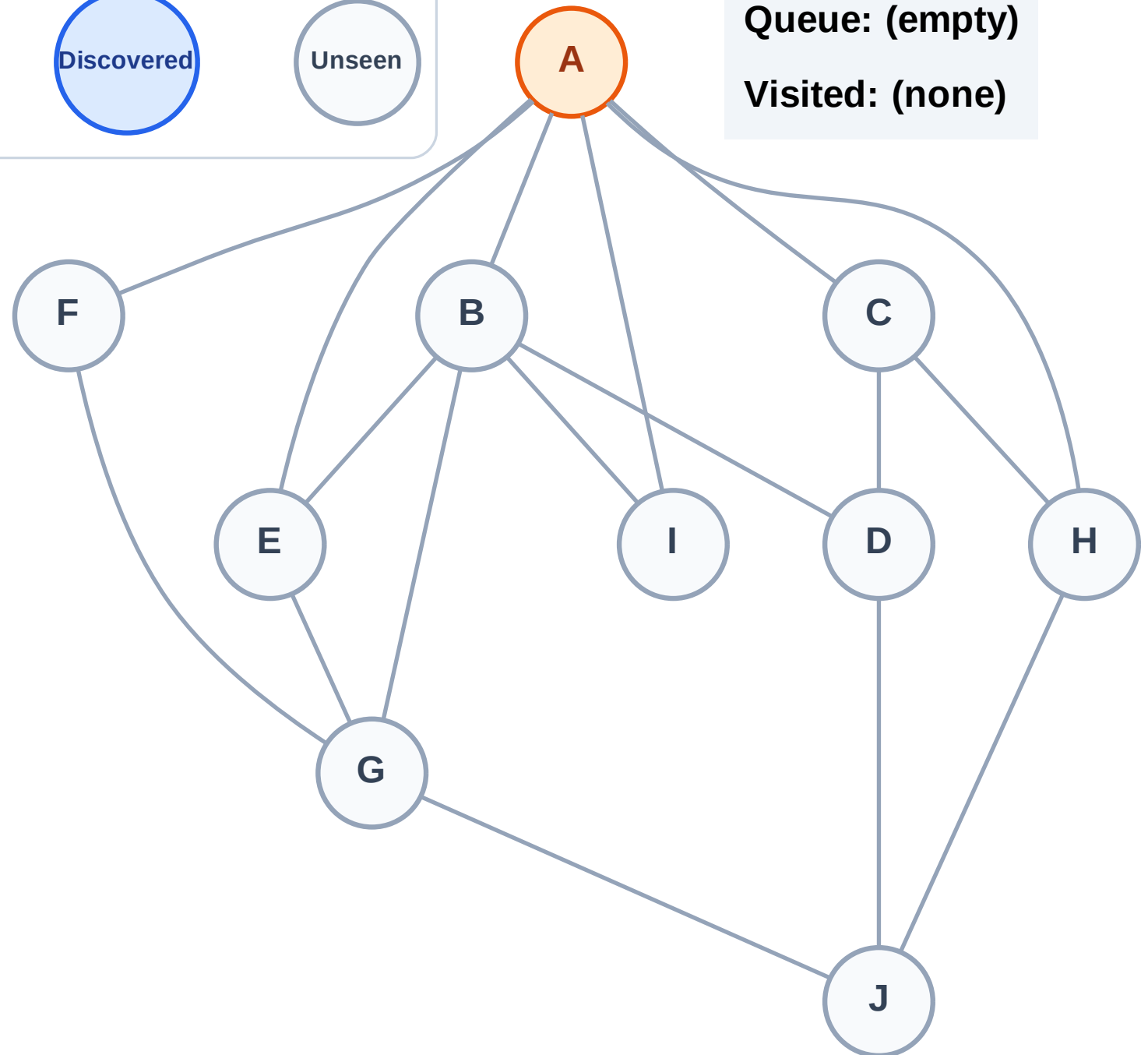
Processing

Discovered

Unseen

Queue: (empty)

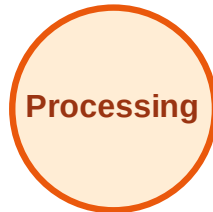
Visited: (none)



BFS Traversal

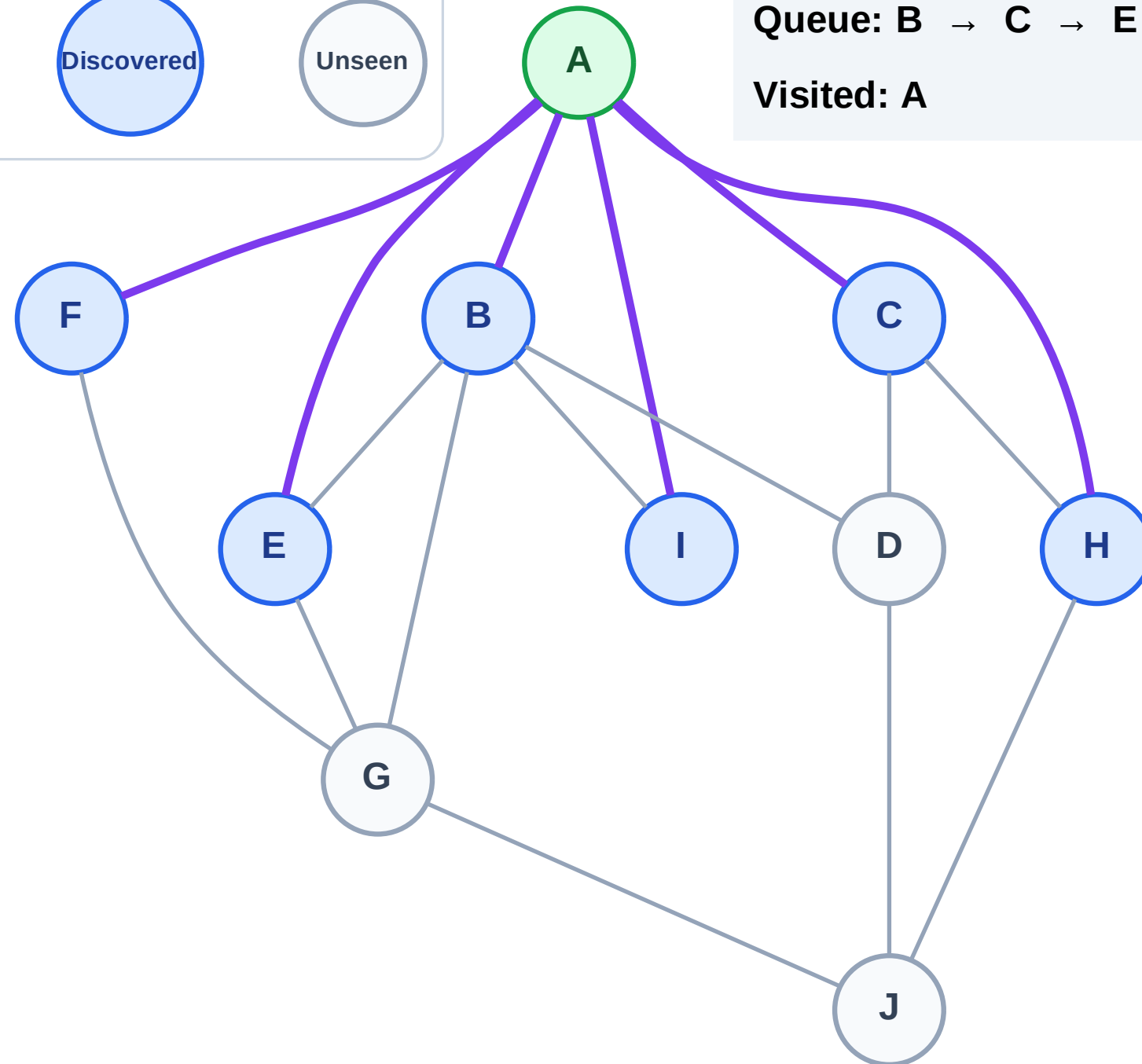
Step 3: Discover B, C, E, F, H, I from A

Legend



Queue: B → C → E → F → H → I

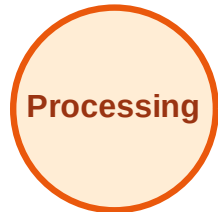
Visited: A



BFS Traversal

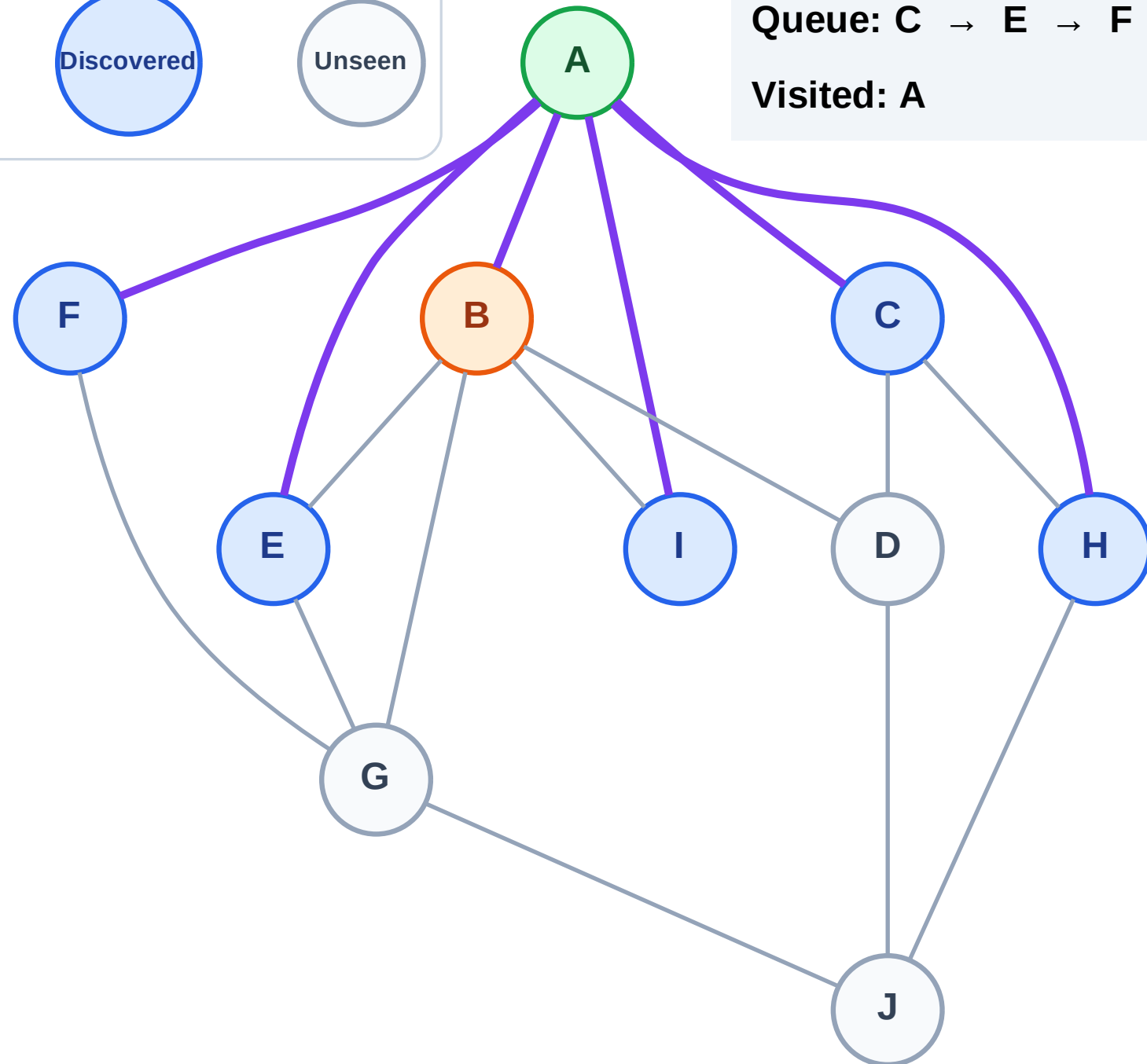
Step 4: Process node B

Legend



Queue: C → E → F → H → I

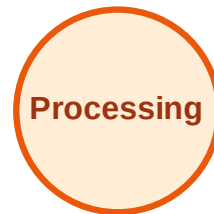
Visited: A



BFS Traversal

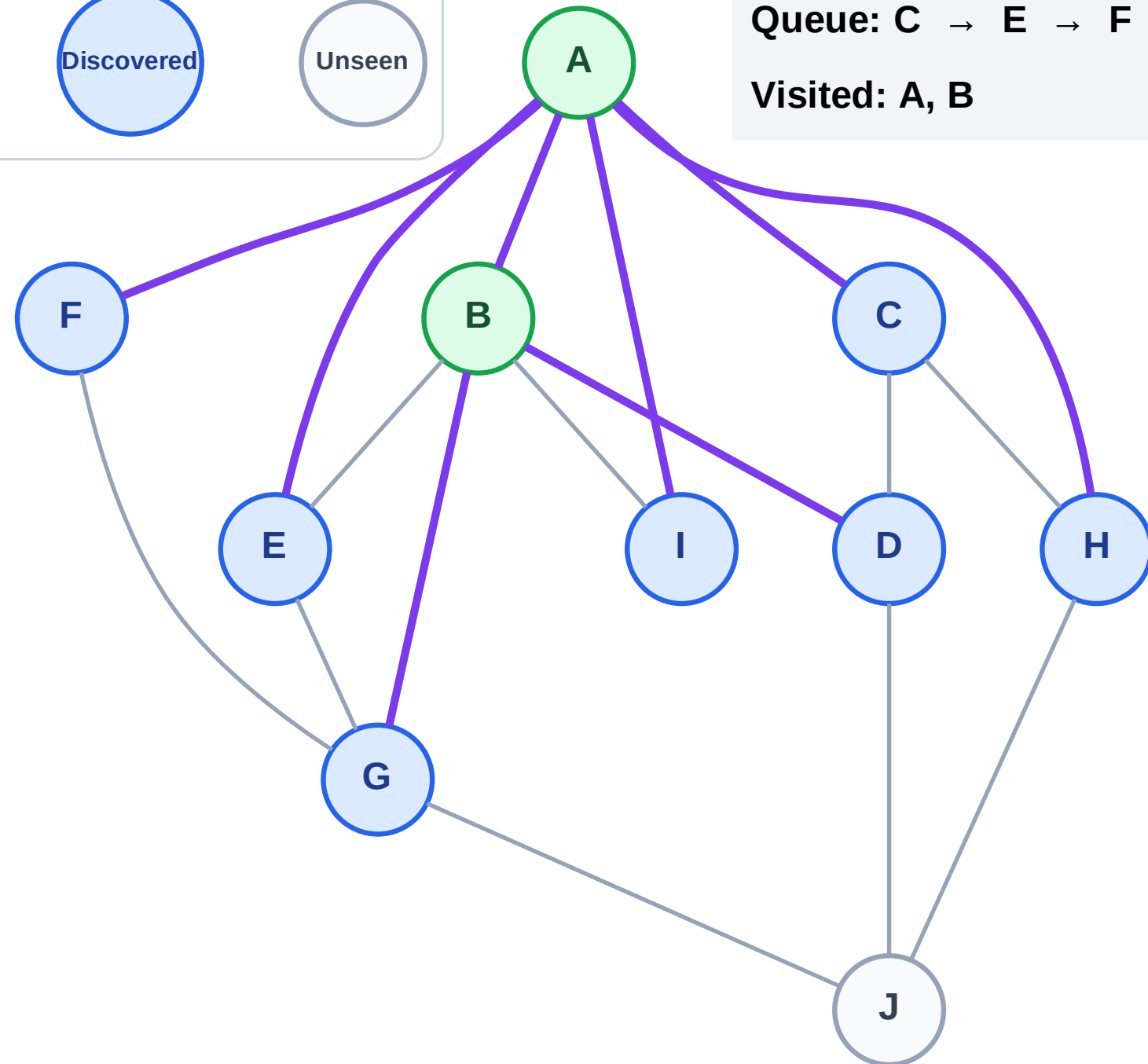
Step 5: Discover D, G from B

Legend



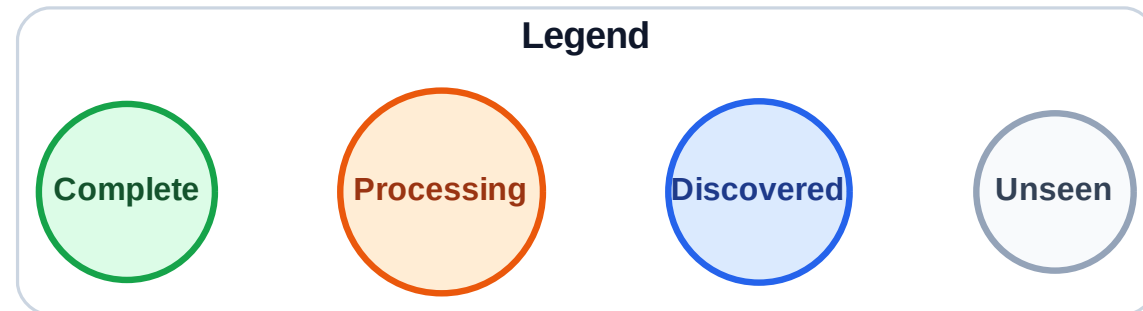
Queue: C → E → F → H → I → D → G

Visited: A, B

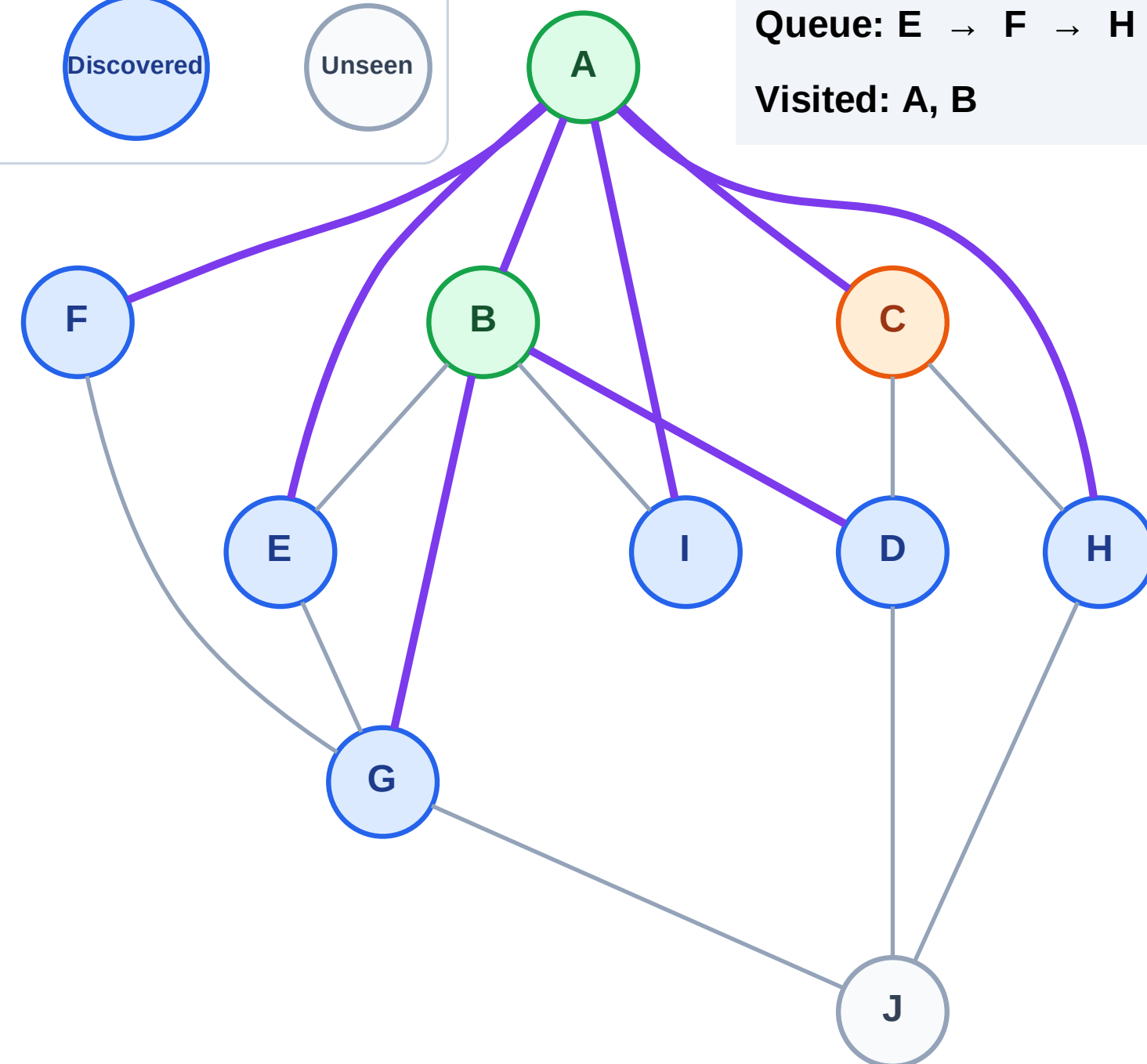


BFS Traversal

Step 6: Process node C



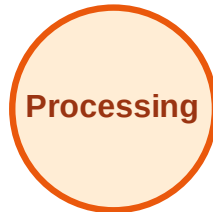
Queue: E → F → H → I → D → G
Visited: A, B



BFS Traversal

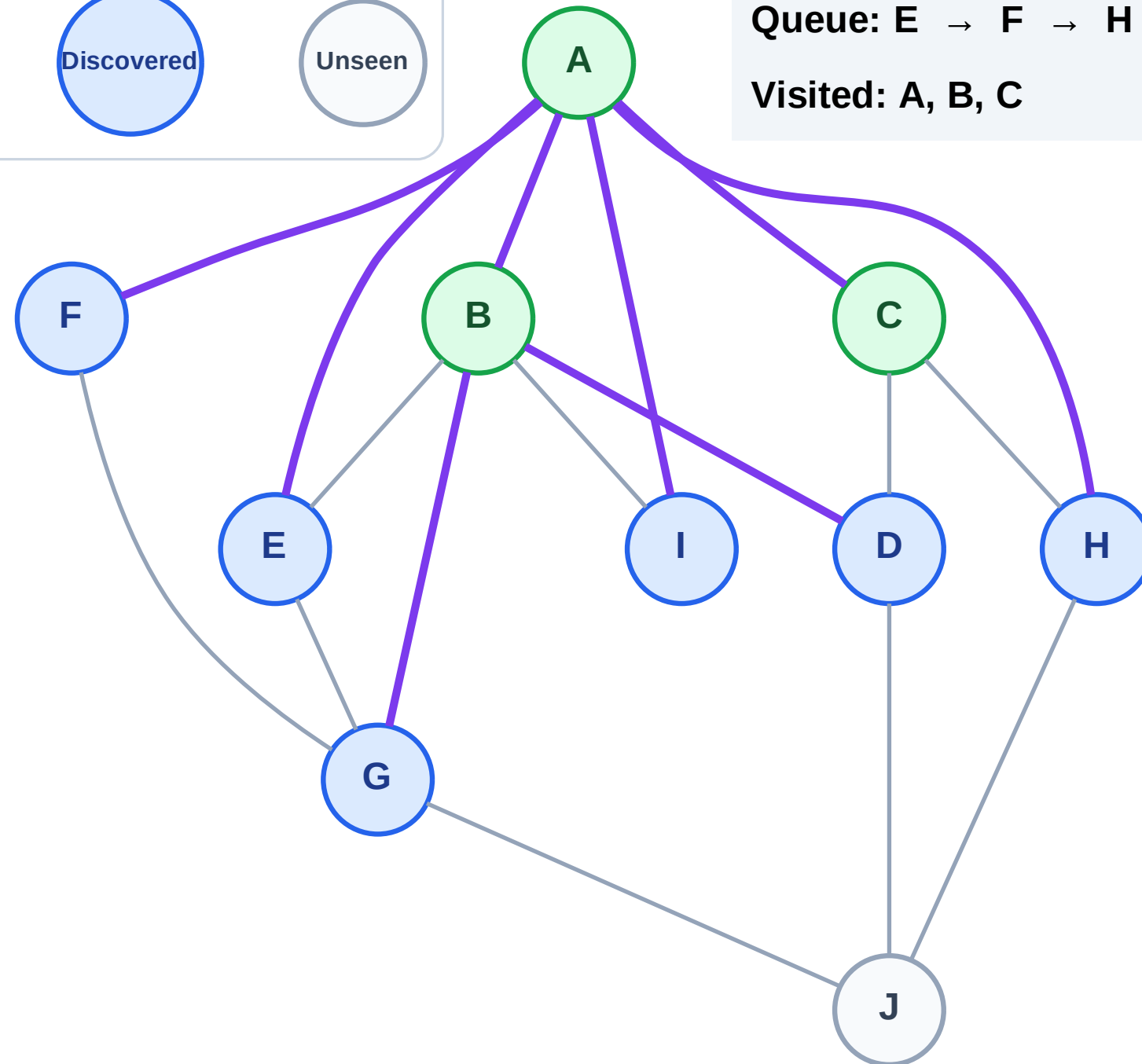
Step 7: No new neighbors from C

Legend



Queue: E → F → H → I → D → G

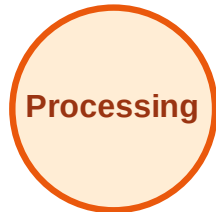
Visited: A, B, C



BFS Traversal

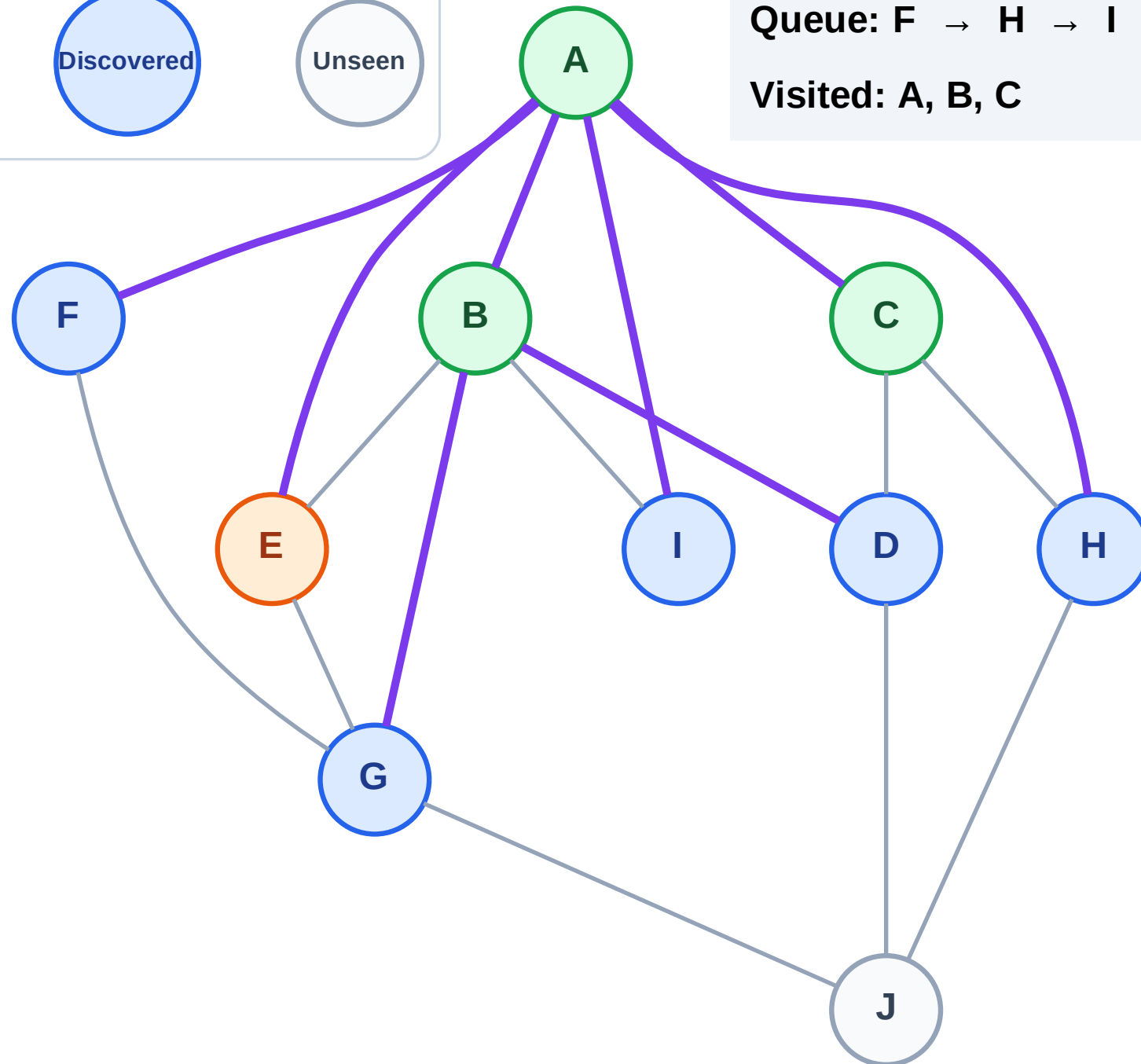
Step 8: Process node E

Legend



Queue: F → H → I → D → G

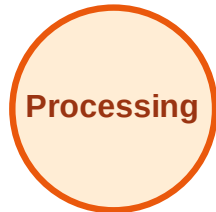
Visited: A, B, C



BFS Traversal

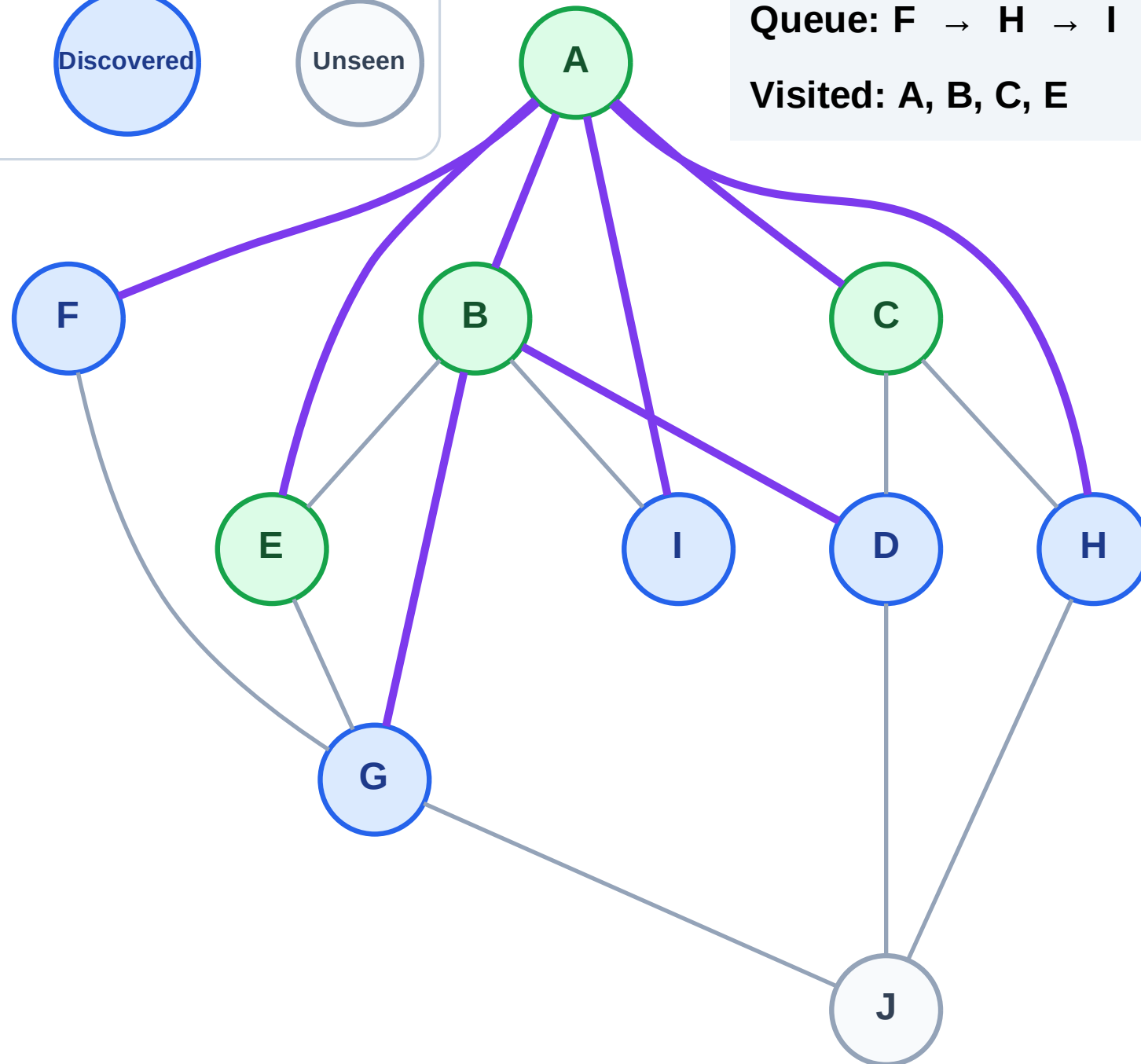
Step 9: No new neighbors from E

Legend



Queue: F → H → I → D → G

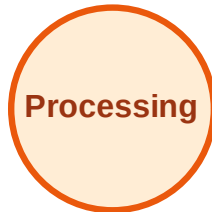
Visited: A, B, C, E



BFS Traversal

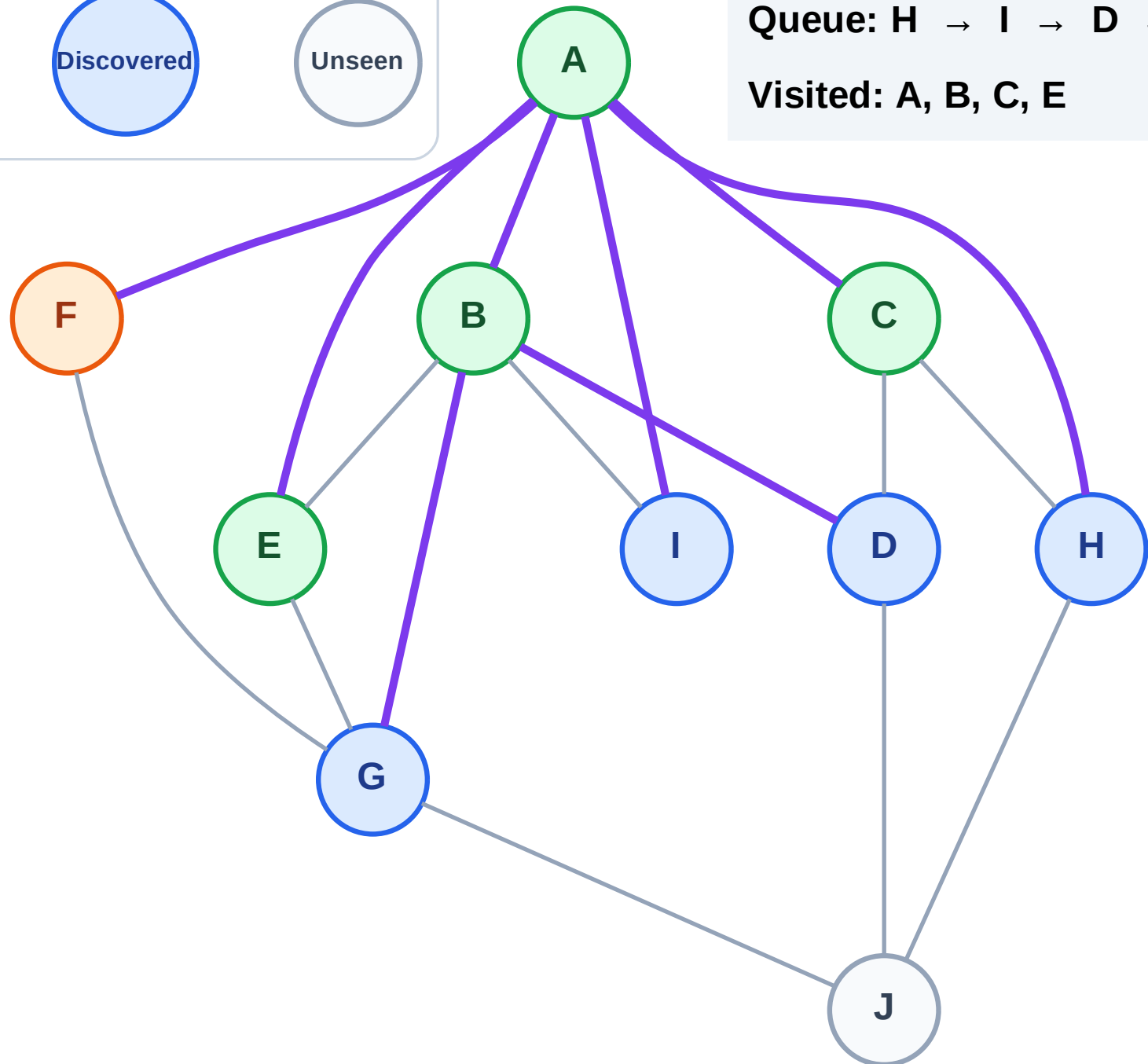
Step 10: Process node F

Legend



Queue: H → I → D → G

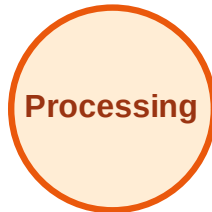
Visited: A, B, C, E



BFS Traversal

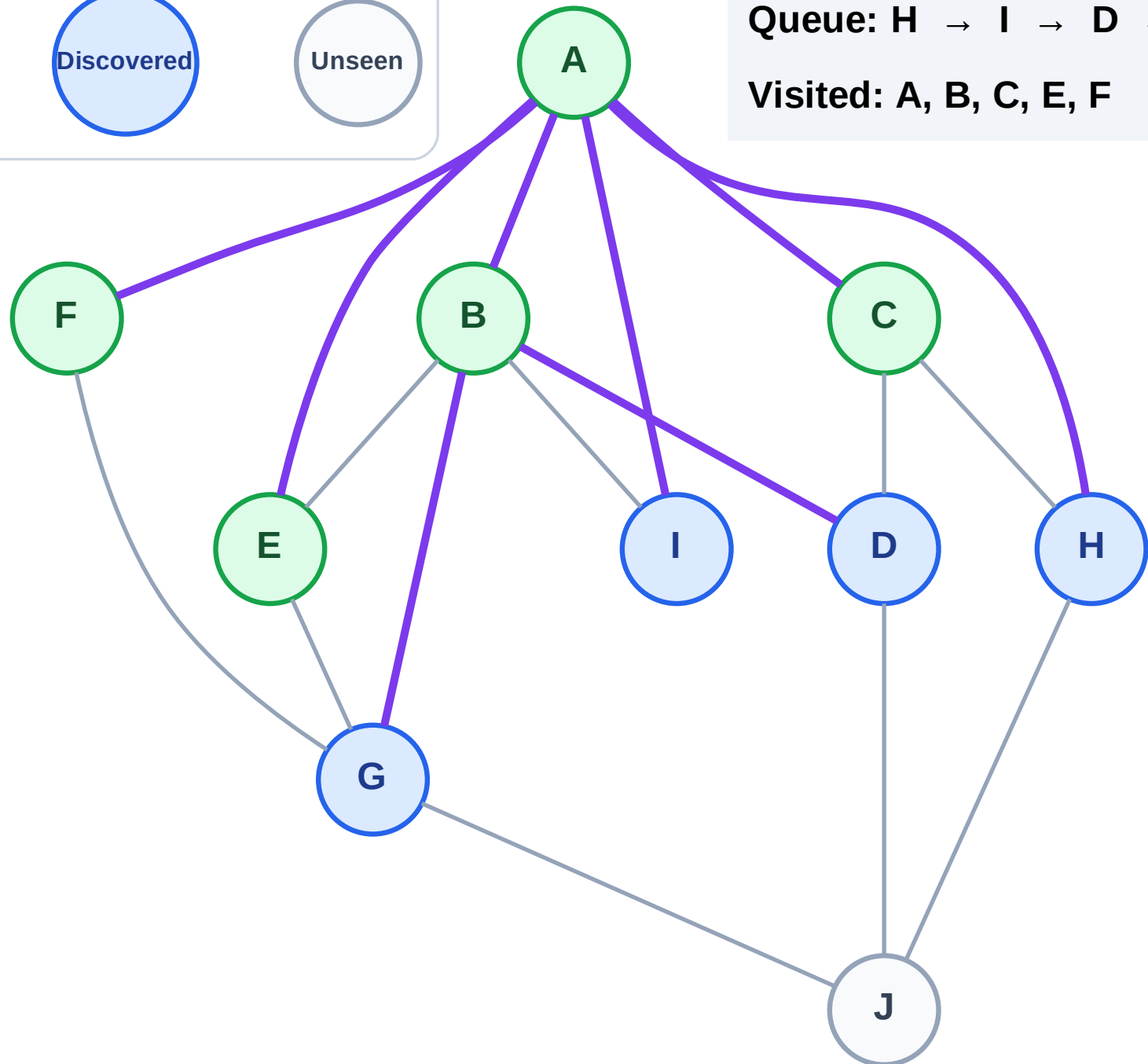
Step 11: No new neighbors from F

Legend



Queue: H → I → D → G

Visited: A, B, C, E, F



BFS Traversal

Step 12: Process node H

Legend

Complete

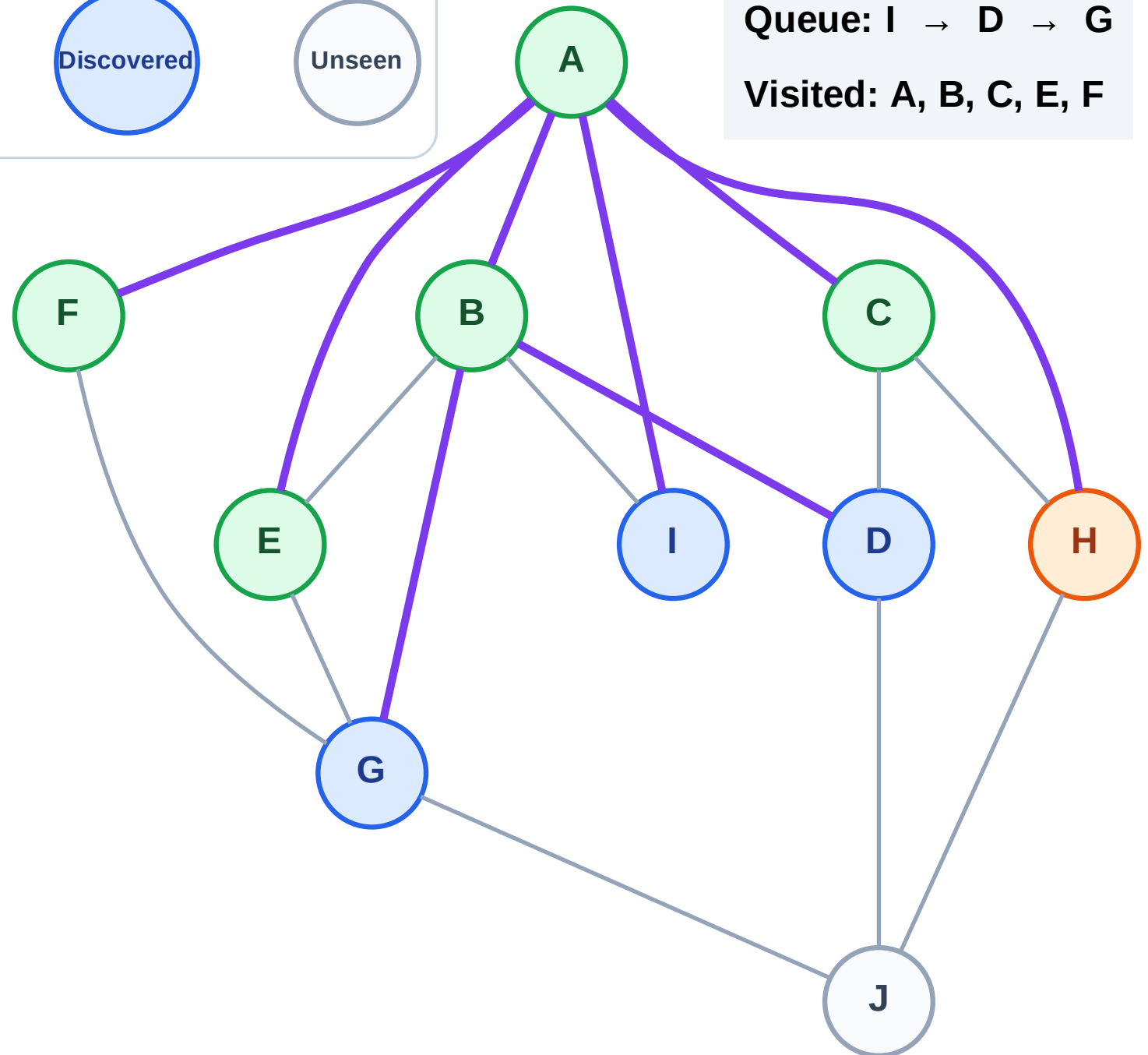
Processing

Discovered

Unseen

Queue: I → D → G

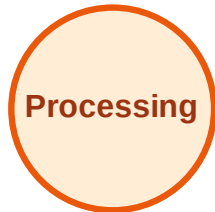
Visited: A, B, C, E, F



BFS Traversal

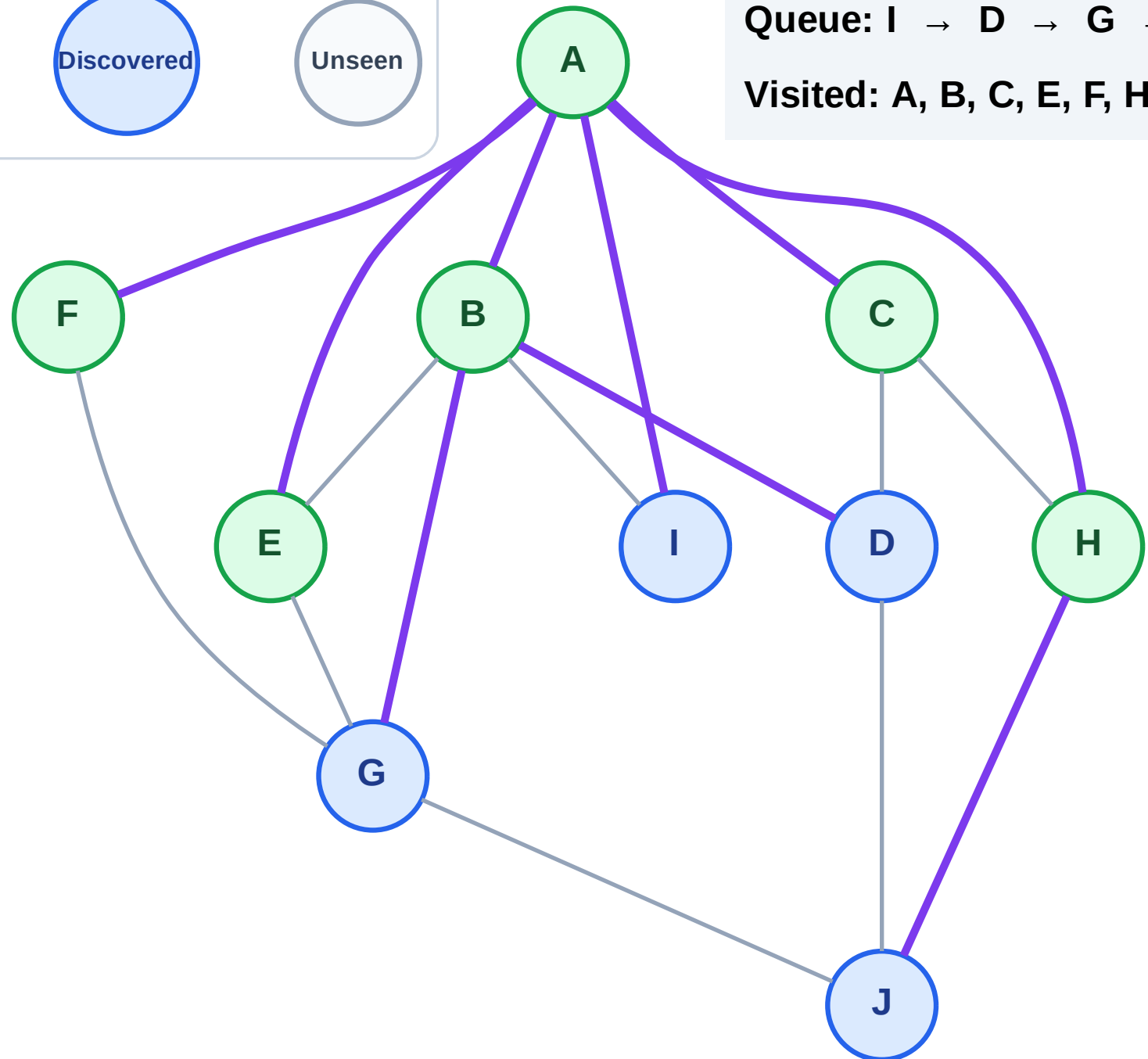
Step 13: Discover J from H

Legend



Queue: I → D → G → J

Visited: A, B, C, E, F, H



BFS Traversal

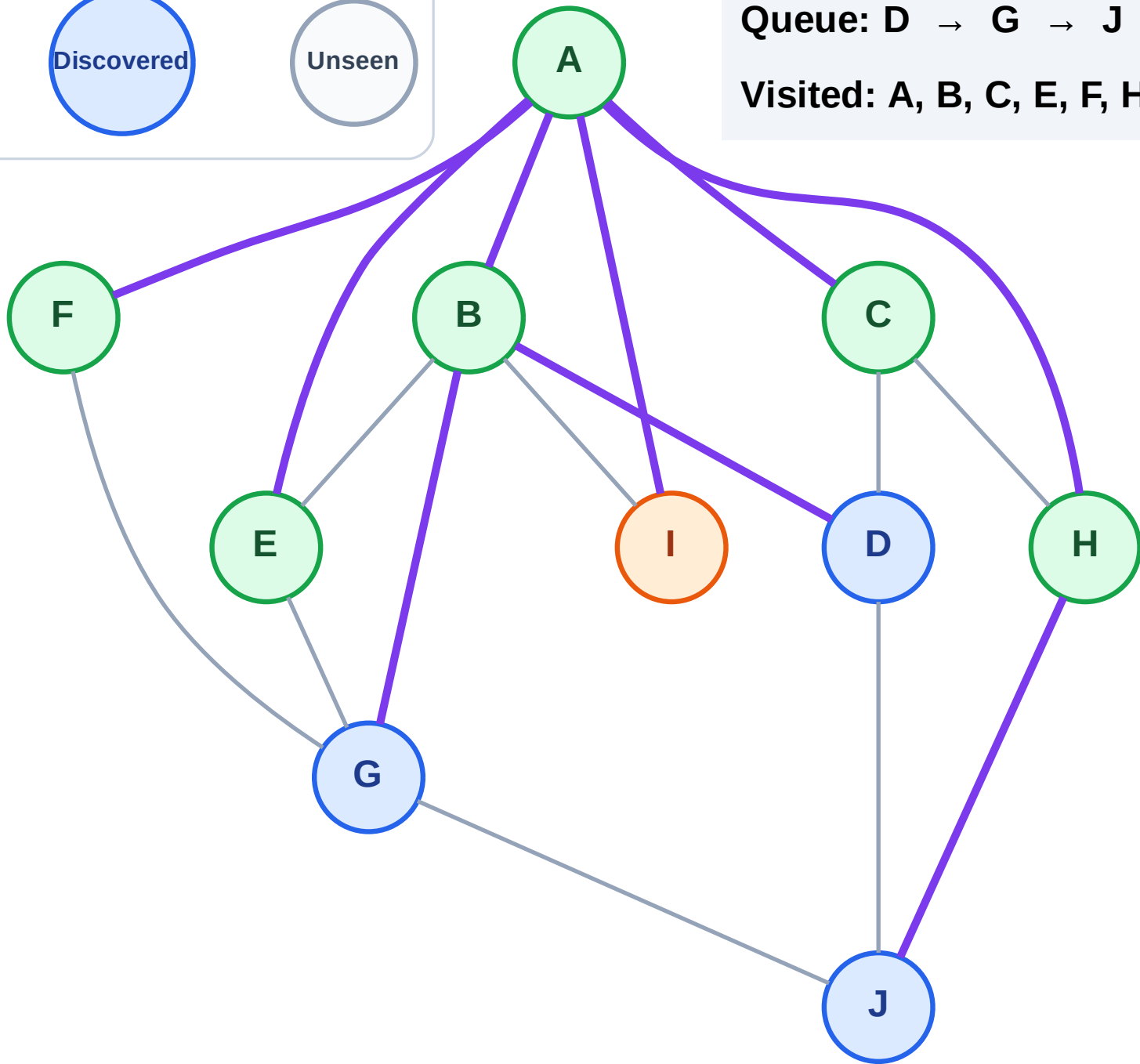
Step 14: Process node I

Legend



Queue: D → G → J

Visited: A, B, C, E, F, H



BFS Traversal

Step 15: No new neighbors from I

Legend

Complete

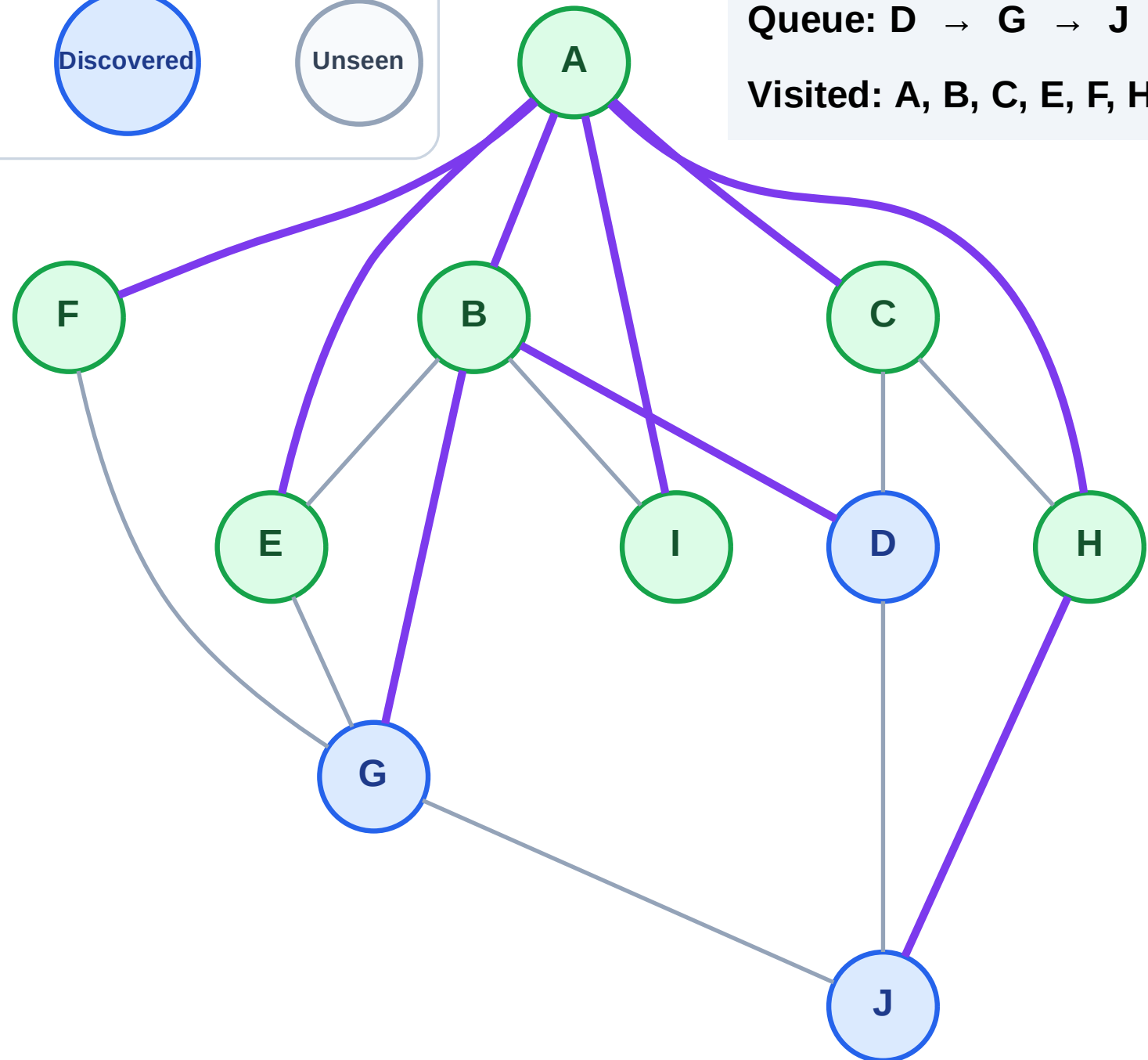
Processing

Discovered

Unseen

Queue: D → G → J

Visited: A, B, C, E, F, H, I



BFS Traversal

Step 16: Process node D

Legend

Complete

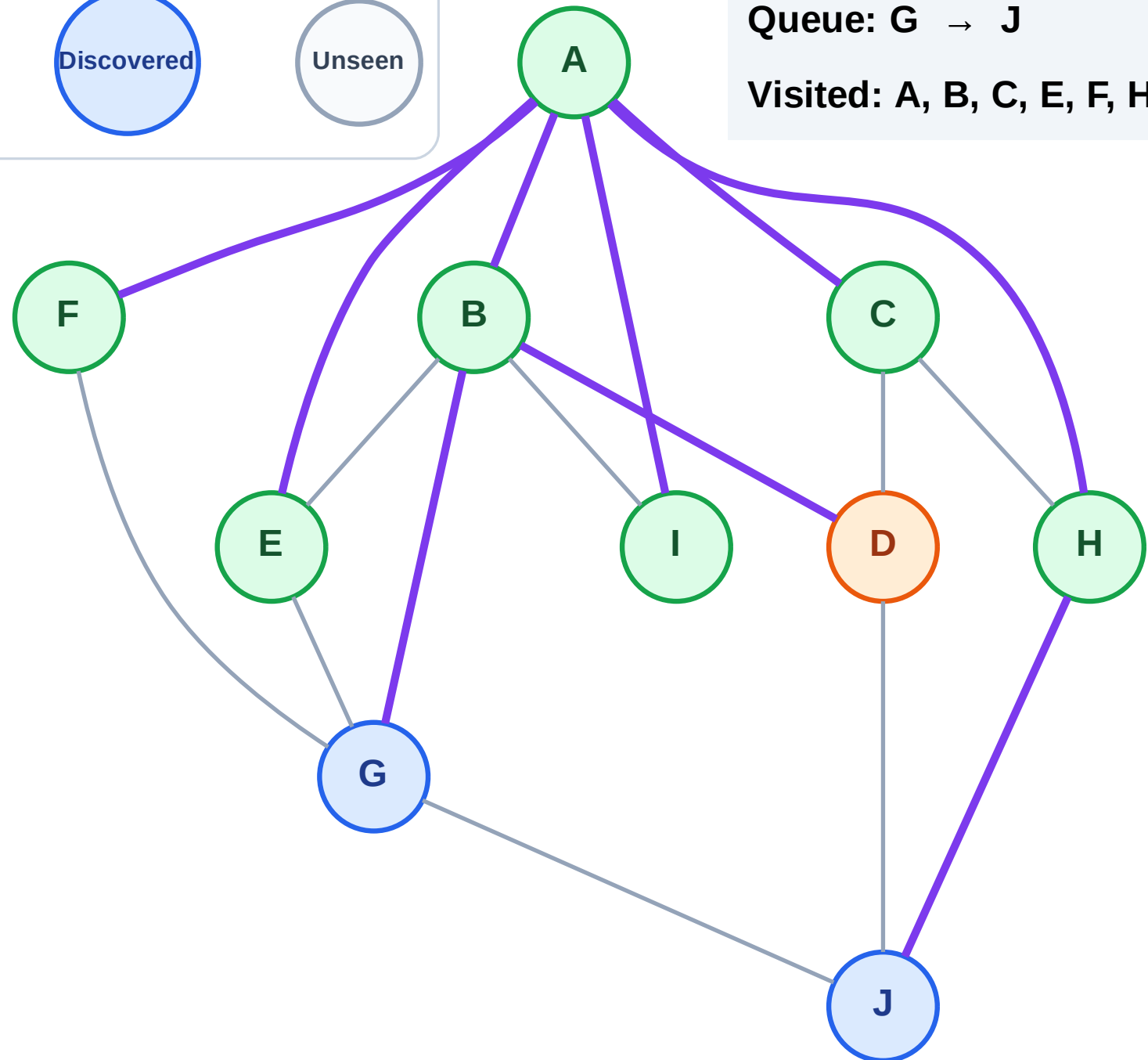
Processing

Discovered

Unseen

Queue: G → J

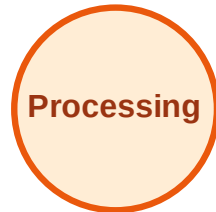
Visited: A, B, C, E, F, H, I



BFS Traversal

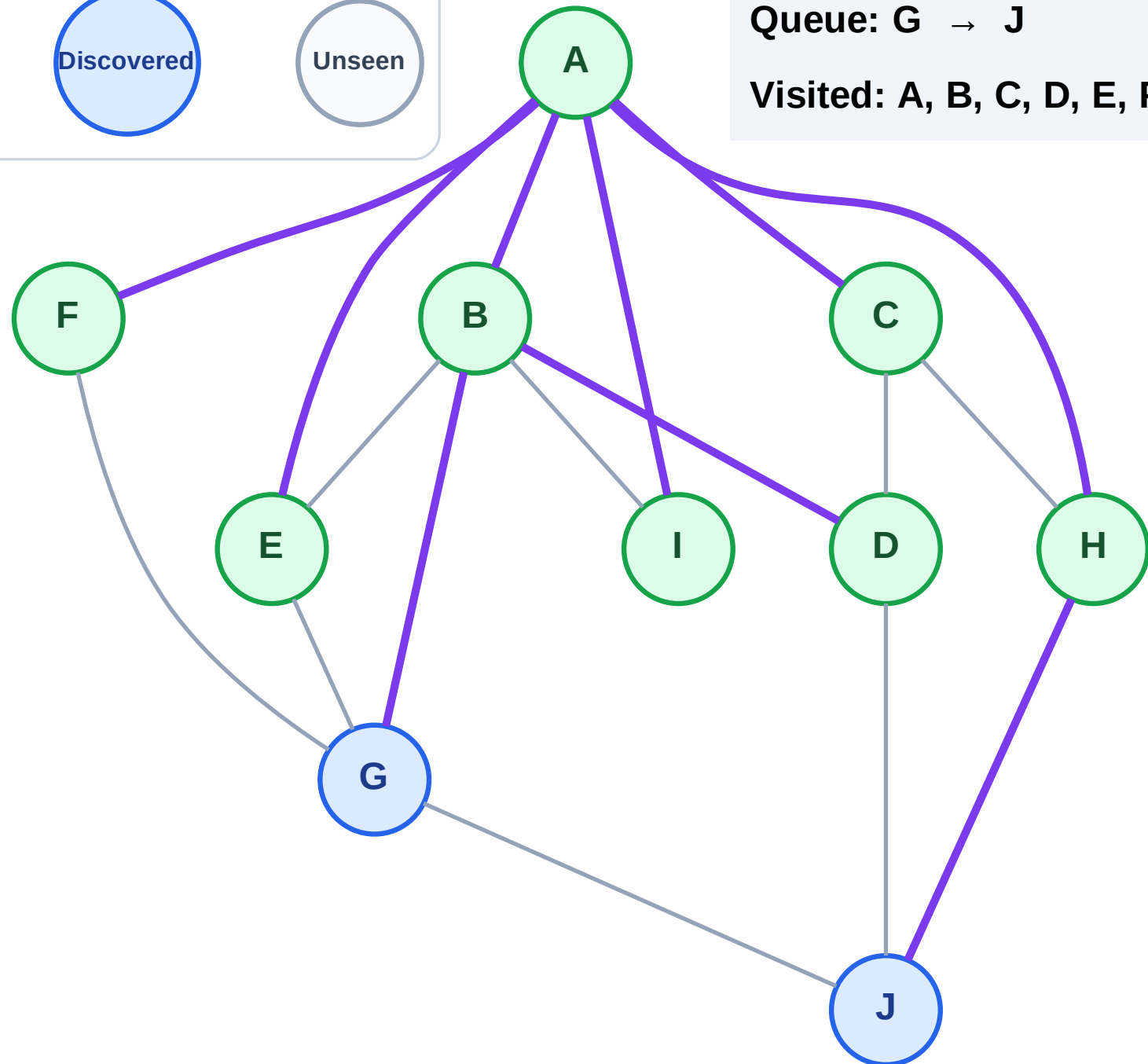
Step 17: No new neighbors from D

Legend



Queue: G → J

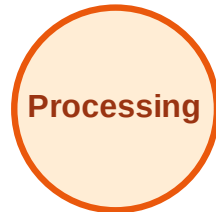
Visited: A, B, C, D, E, F, H, I



BFS Traversal

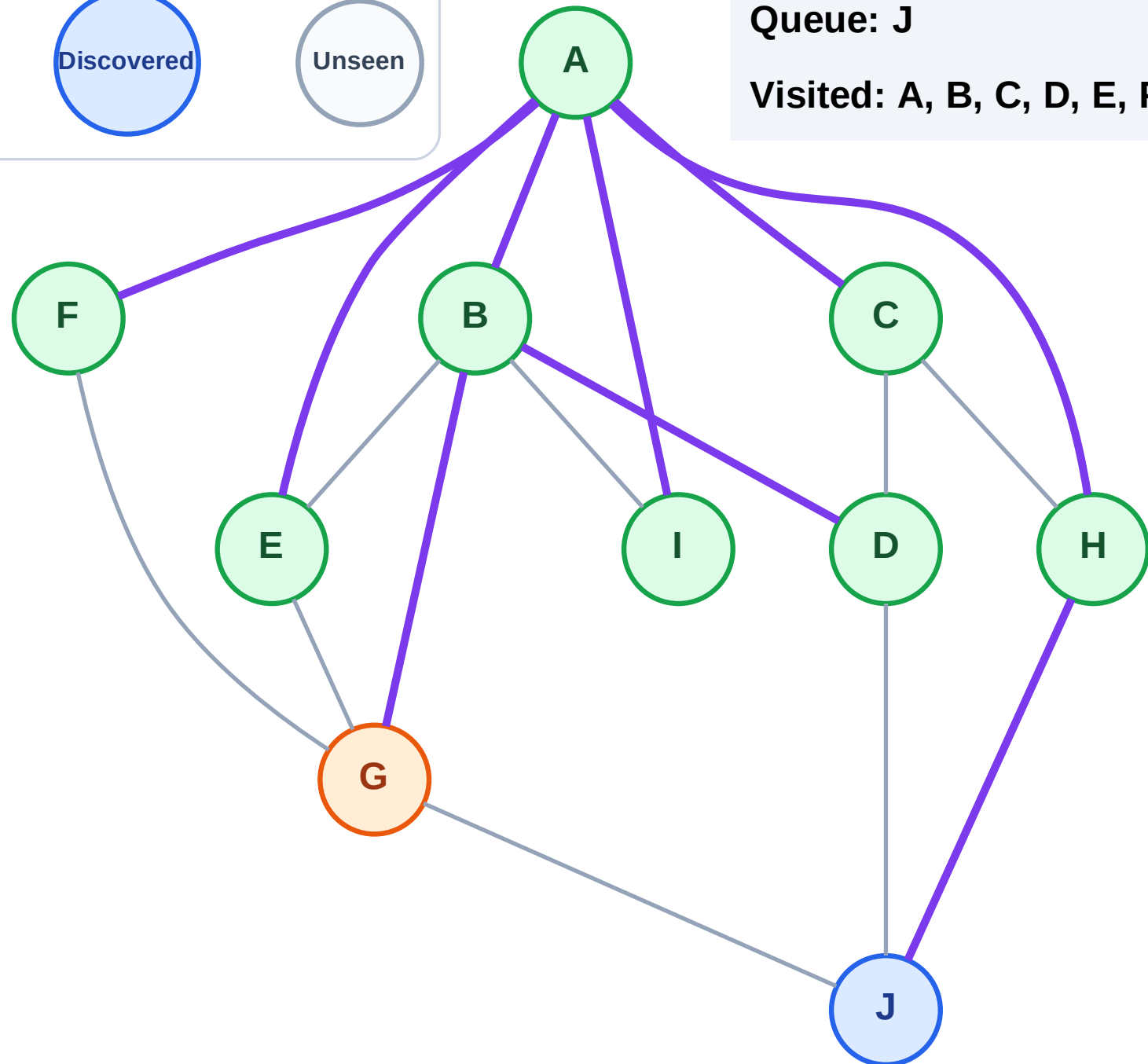
Step 18: Process node G

Legend



Queue: J

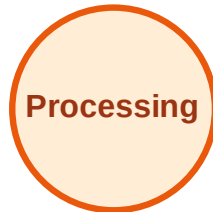
Visited: A, B, C, D, E, F, H, I



BFS Traversal

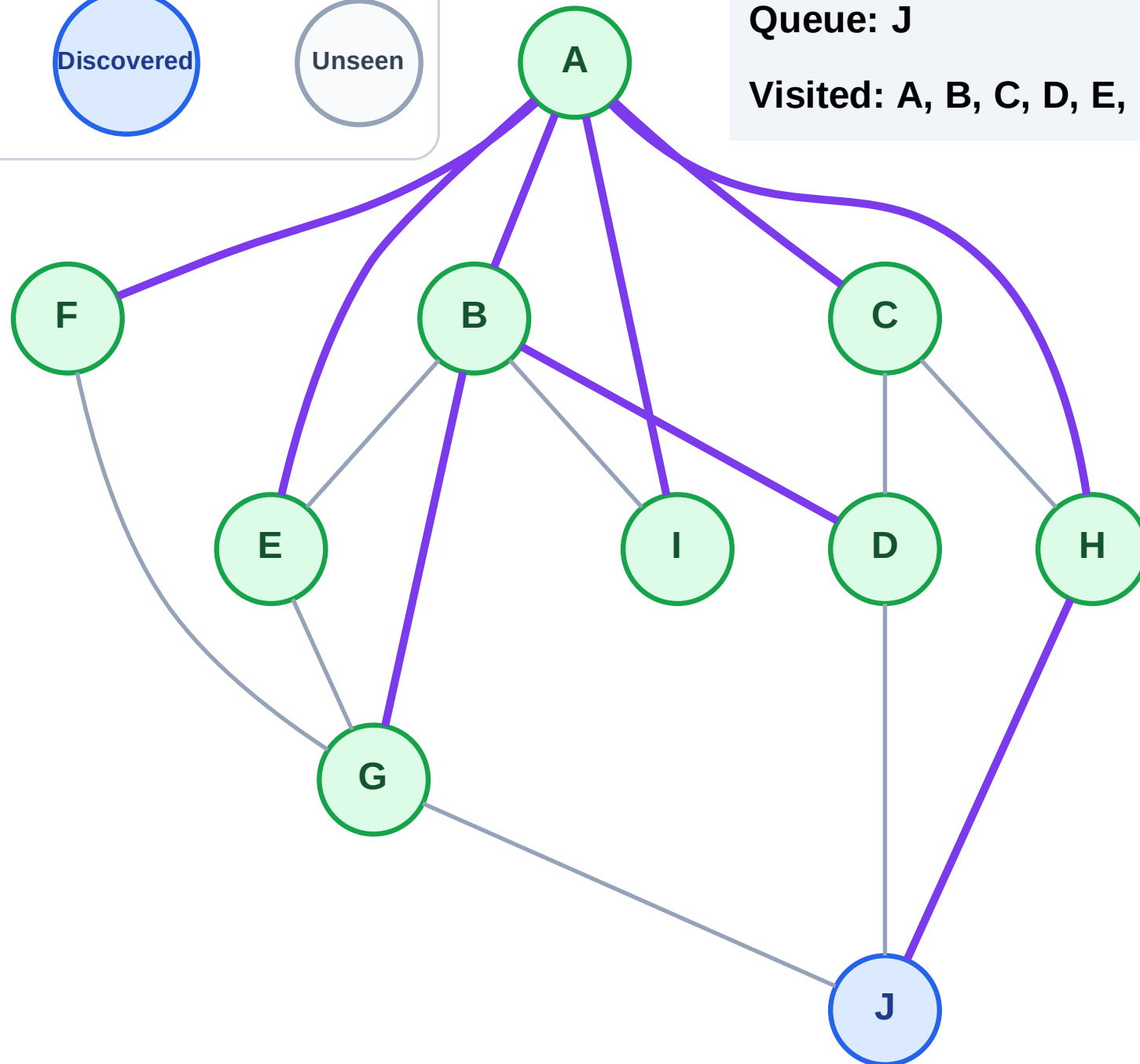
Step 19: No new neighbors from G

Legend



Queue: J

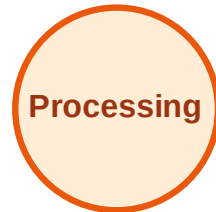
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

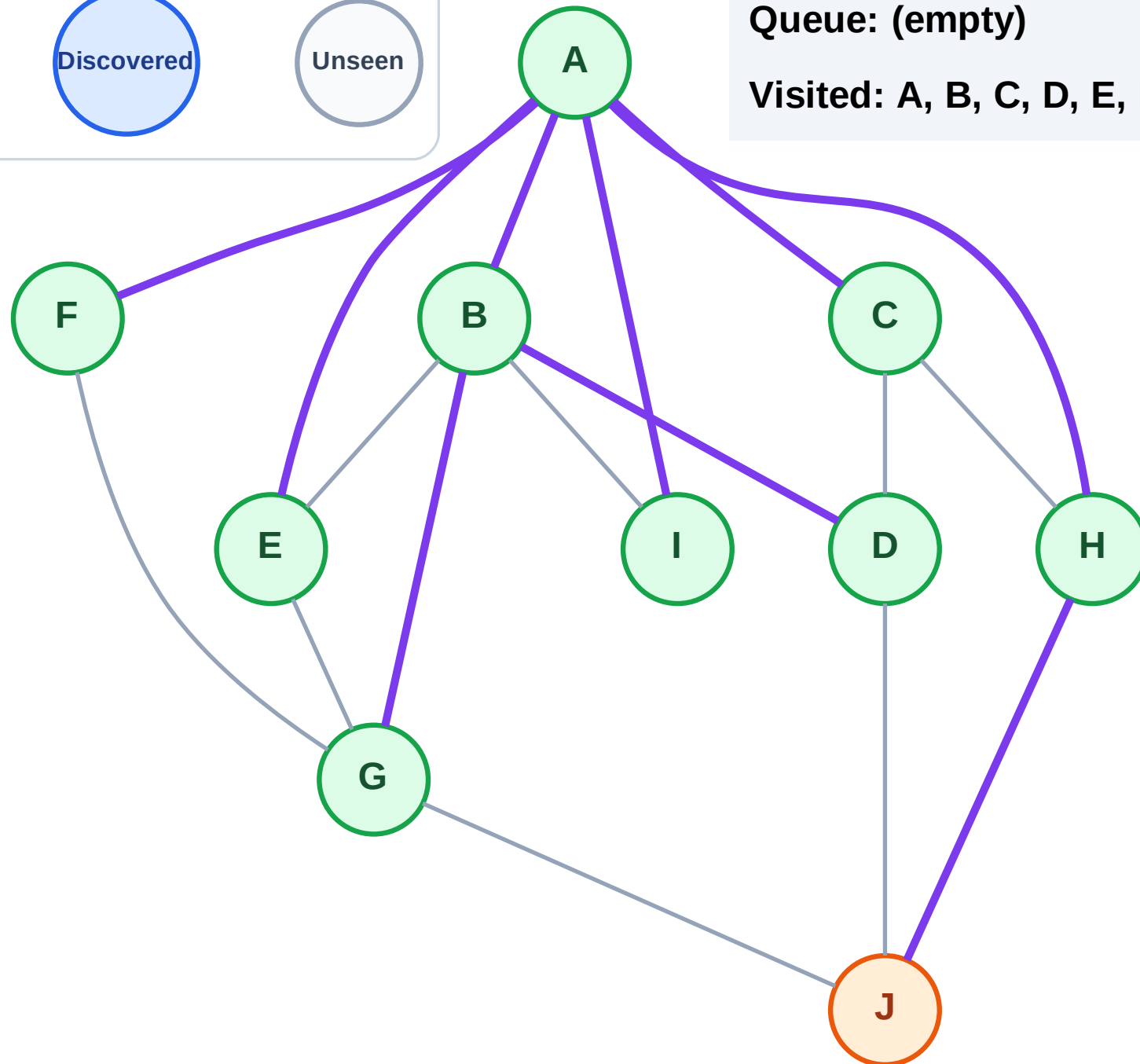
Step 20: Process node J

Legend



Queue: (empty)

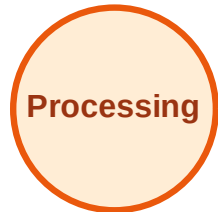
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

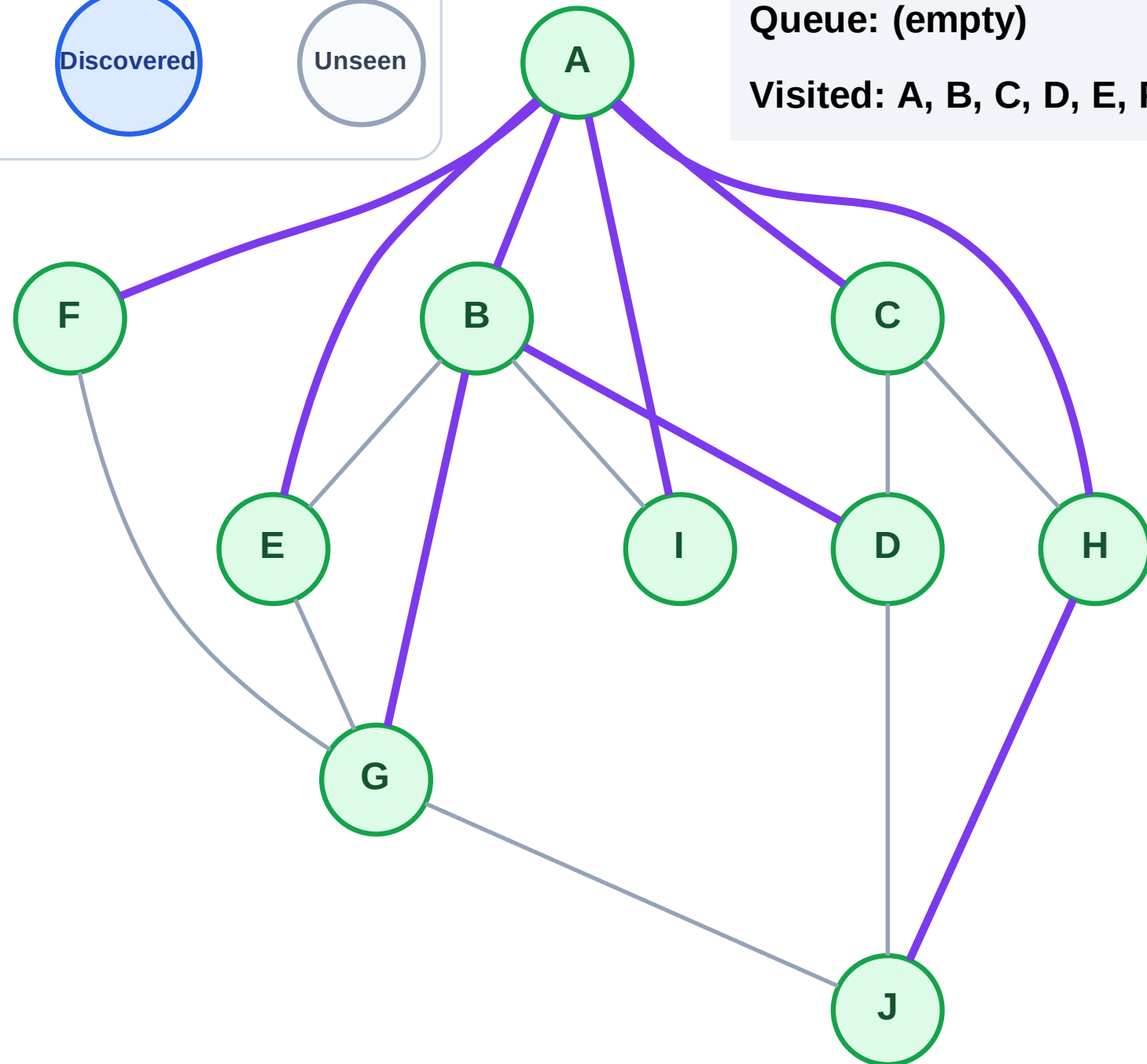
Step 21: No new neighbors from J

Legend



Queue: (empty)

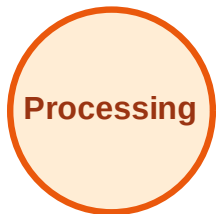
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

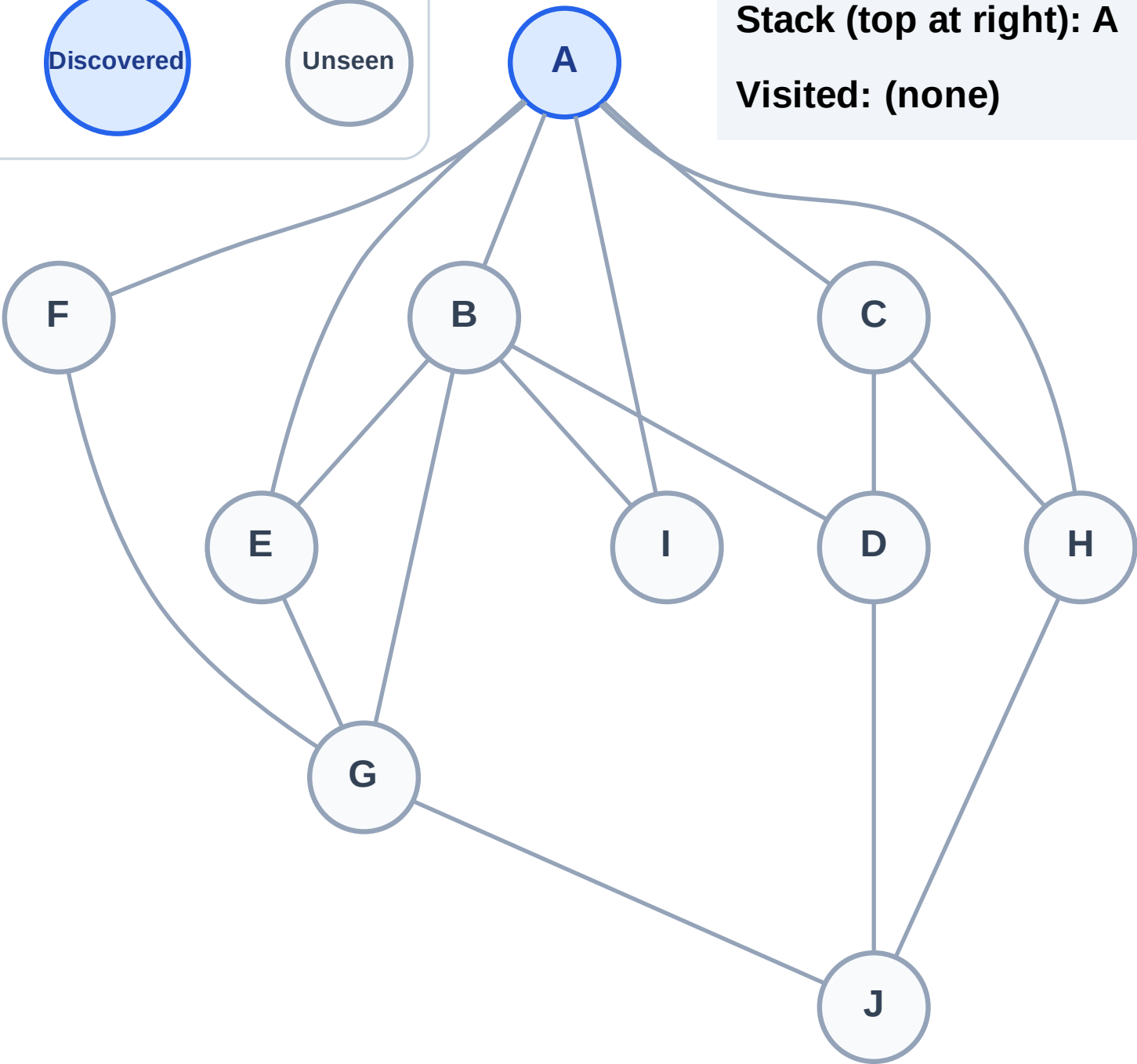
Step 1: Start at A

Legend



Stack (top at right): A

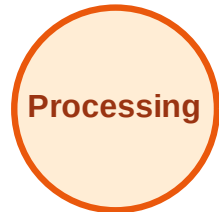
Visited: (none)



DFS Traversal

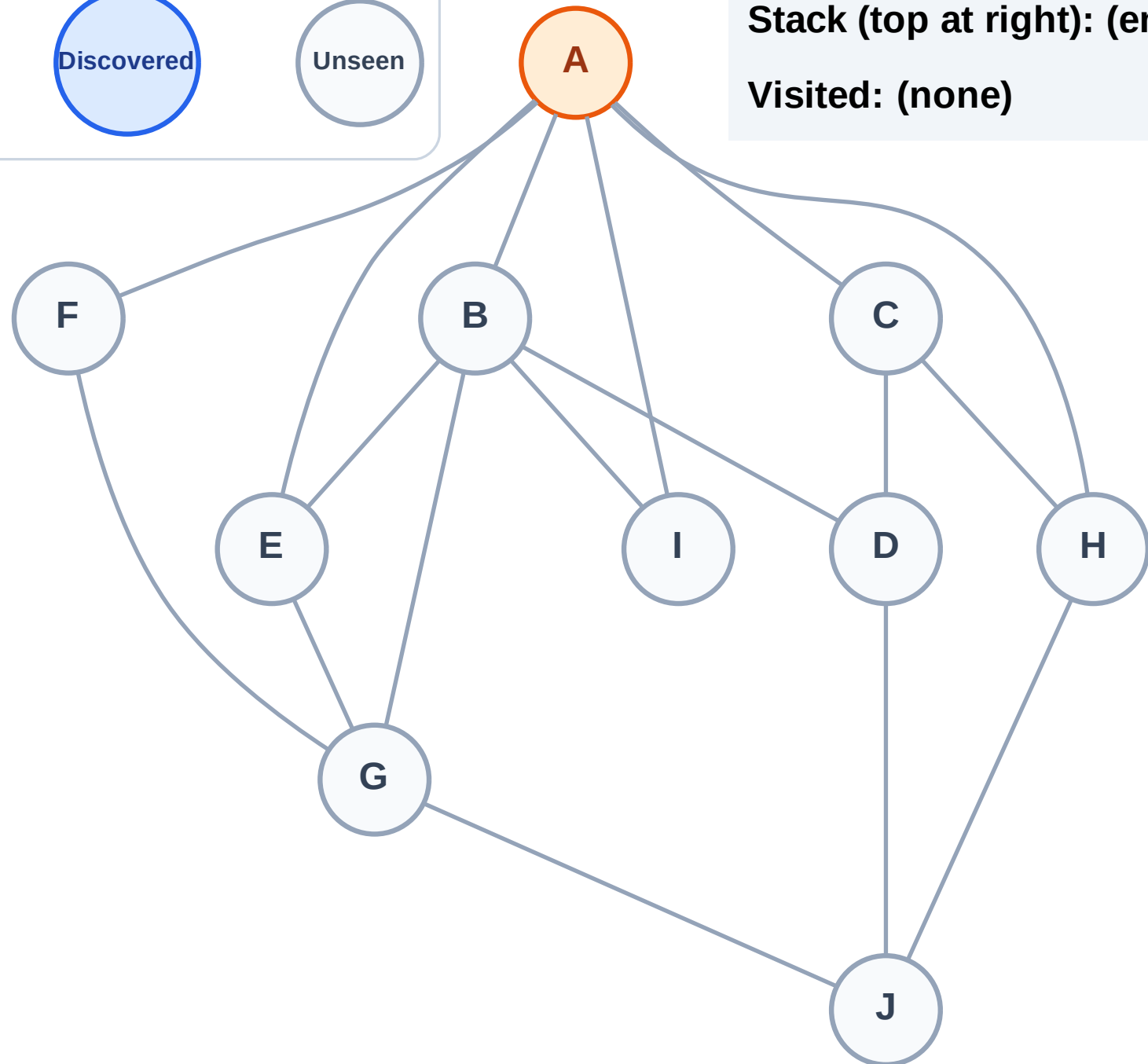
Step 2: Process node A

Legend



Stack (top at right): (empty)

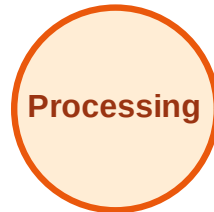
Visited: (none)



DFS Traversal

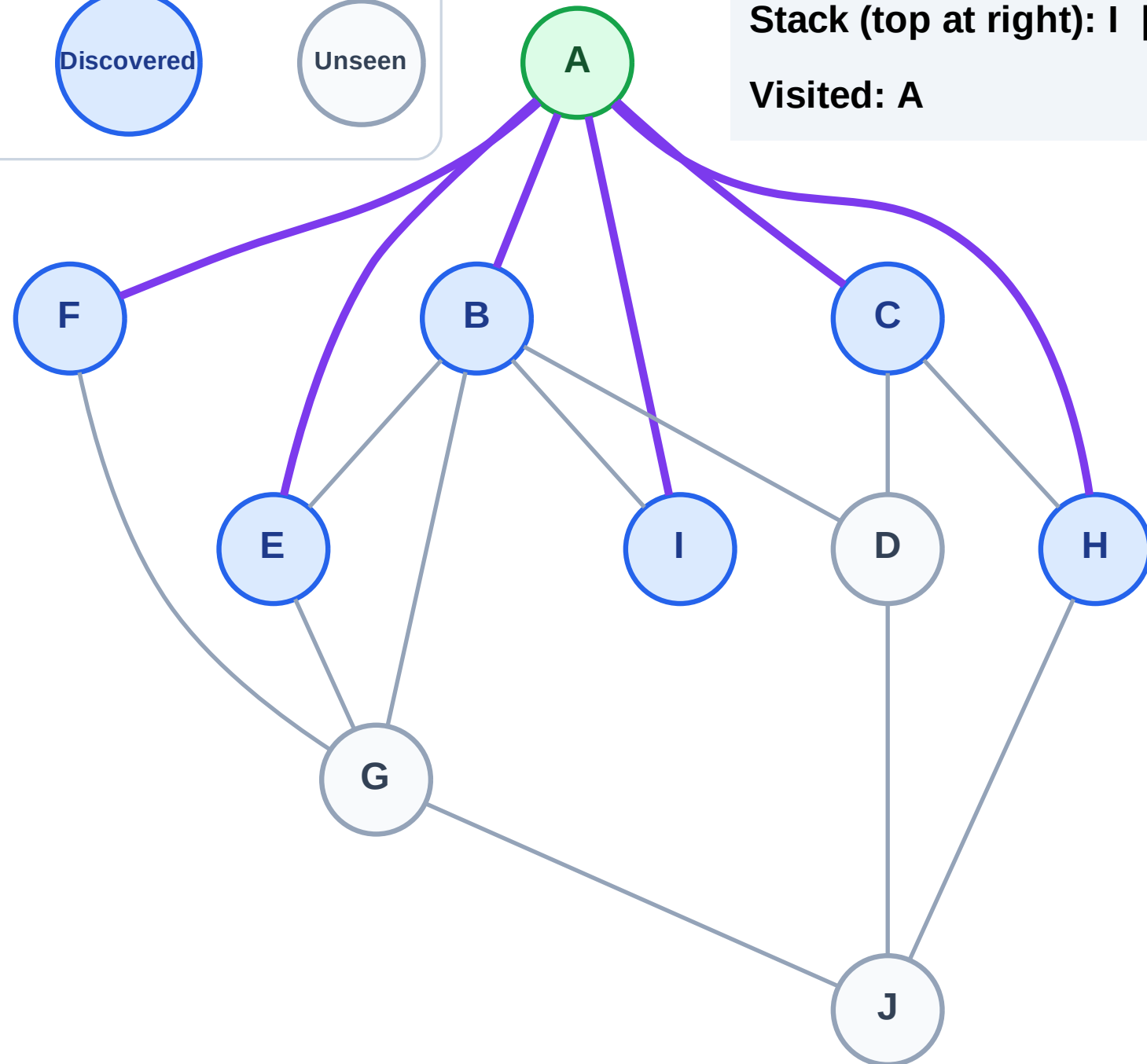
Step 3: Discover I, H, F, E, C, B from A

Legend



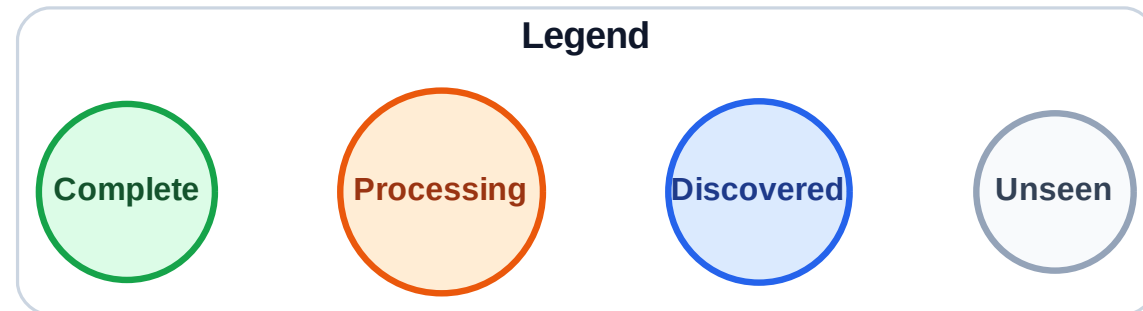
Stack (top at right): I | H | F | E | C | B

Visited: A

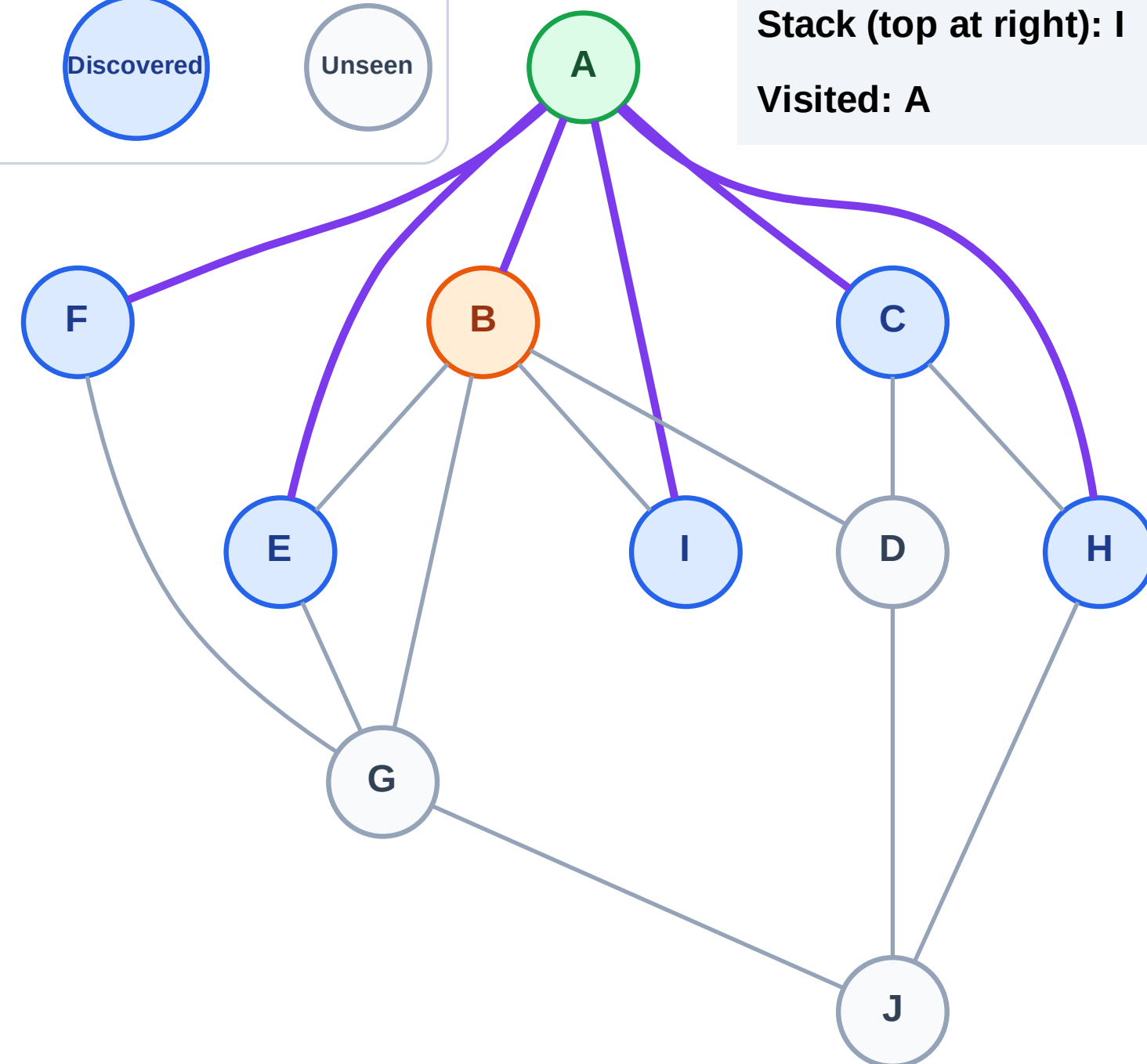


DFS Traversal

Step 4: Process node B



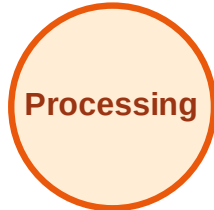
Stack (top at right): I | H | F | E | C
Visited: A



DFS Traversal

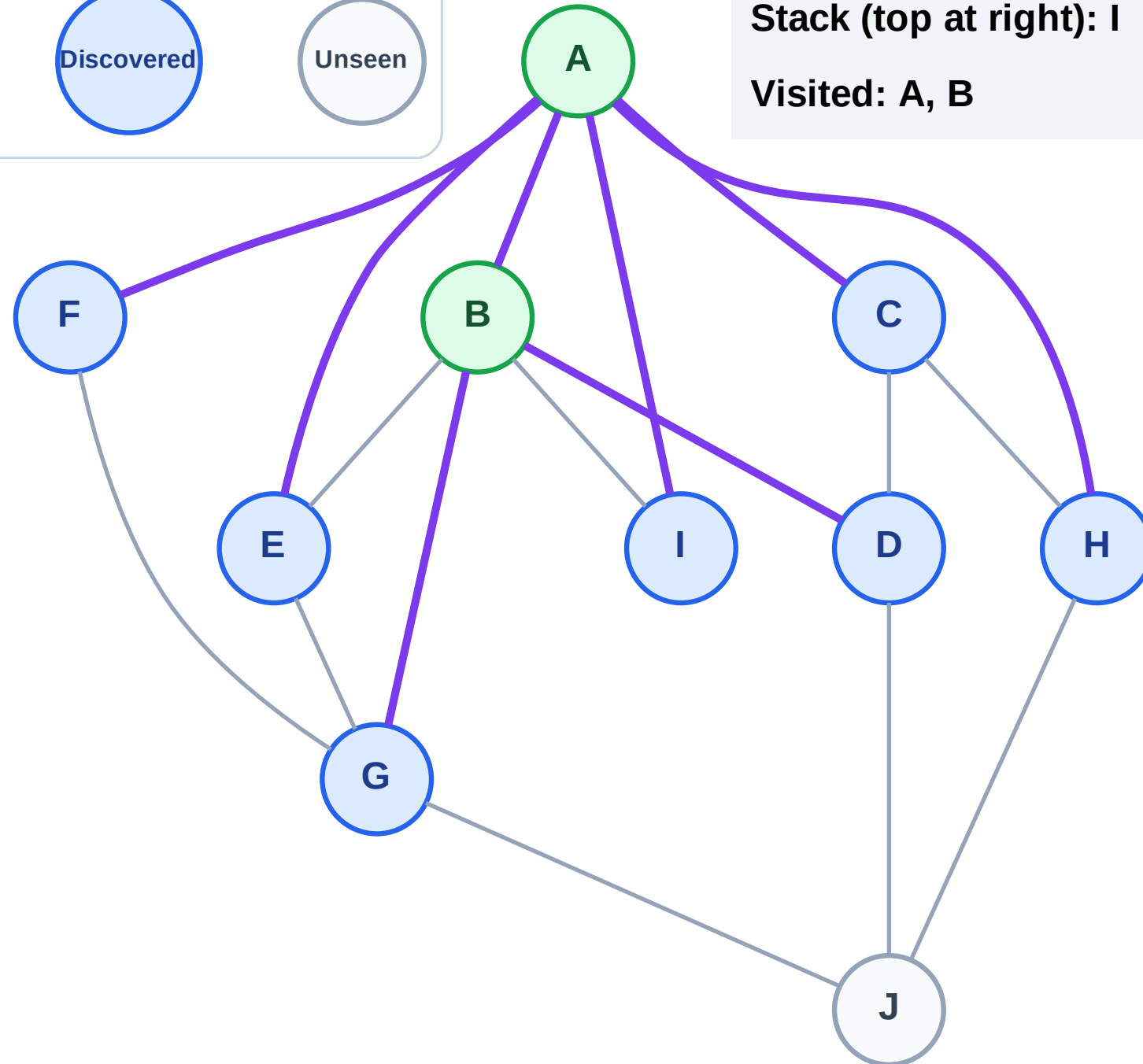
Step 5: Discover G, D from B

Legend



Stack (top at right): I | H | F | E | C | G | D

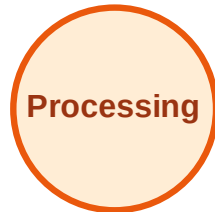
Visited: A, B



DFS Traversal

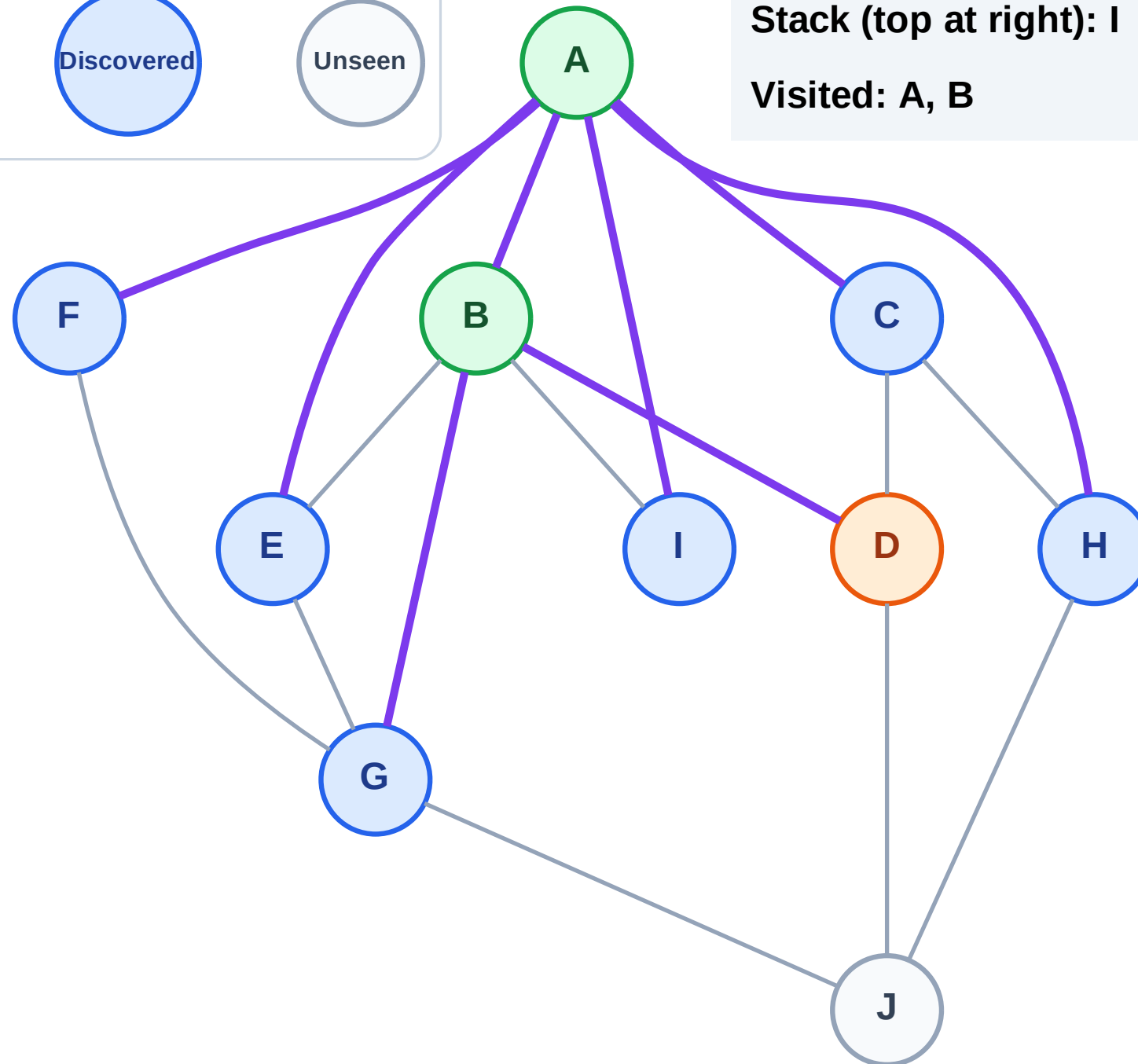
Step 6: Process node D

Legend



Stack (top at right): I | H | F | E | C | G

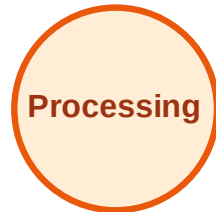
Visited: A, B



DFS Traversal

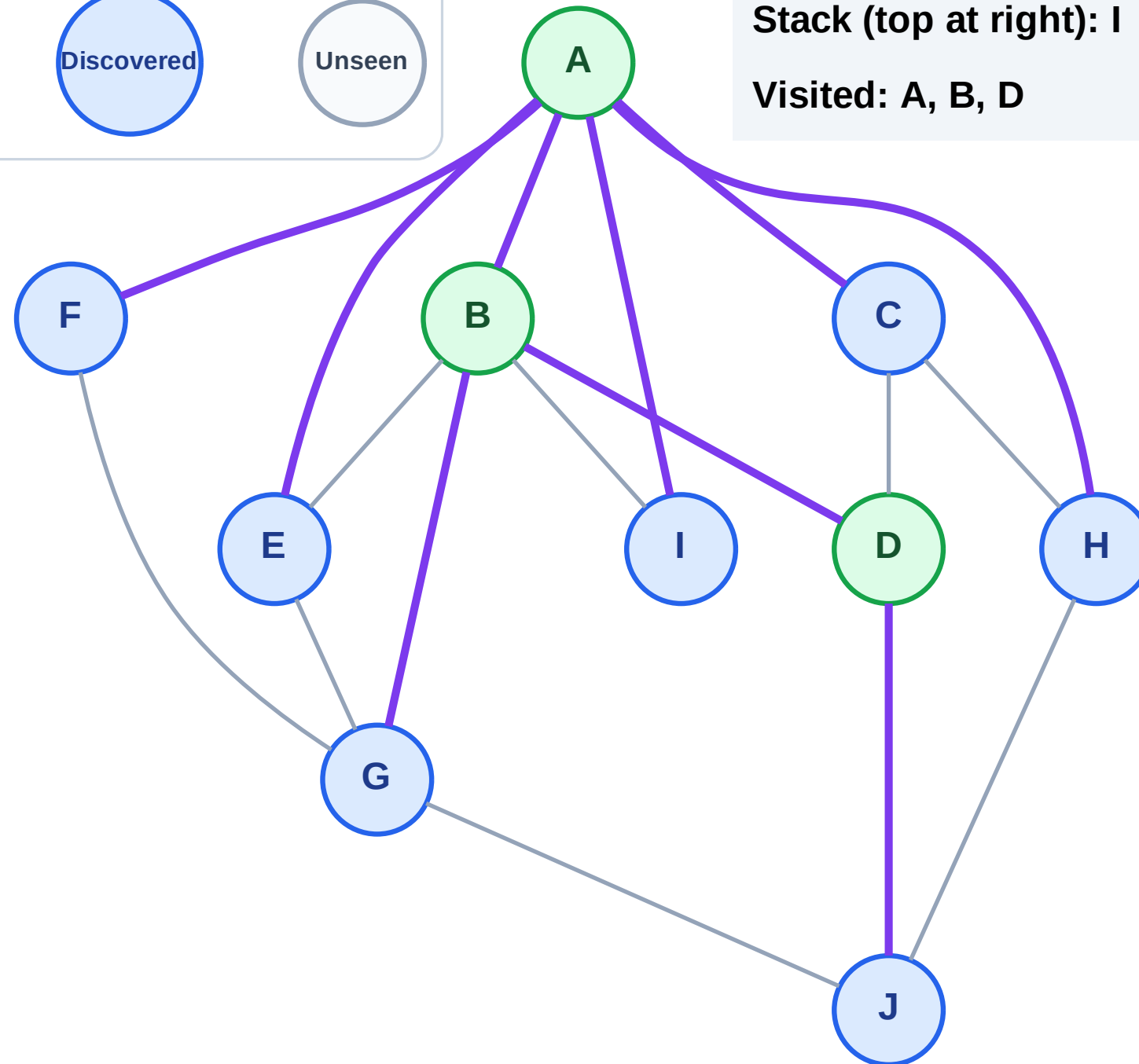
Step 7: Discover J from D

Legend



Stack (top at right): I | H | F | E | C | G | J

Visited: A, B, D



DFS Traversal

Step 8: Process node J

Legend

Complete

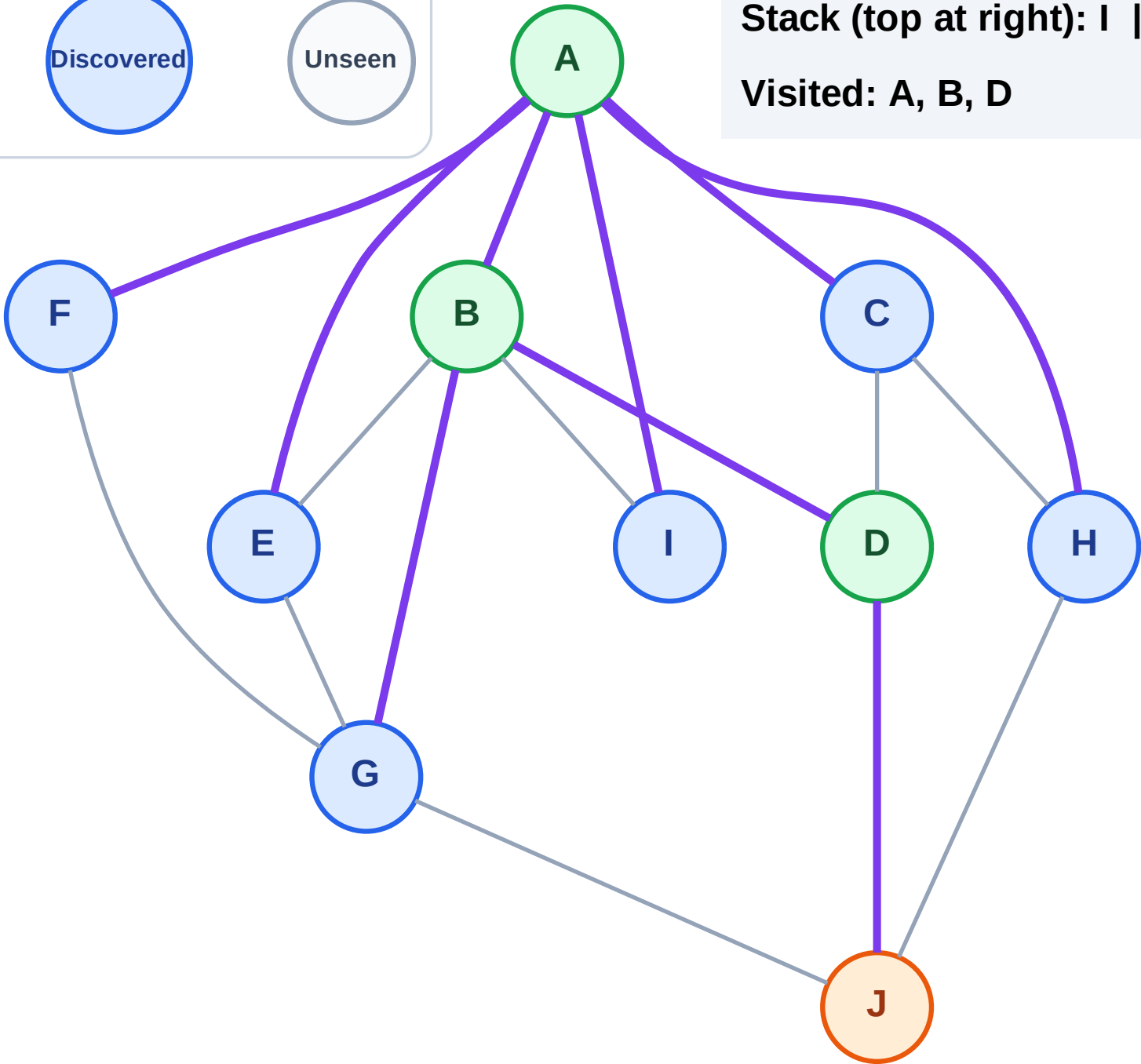
Processing

Discovered

Unseen

Stack (top at right): I | H | F | E | C | G

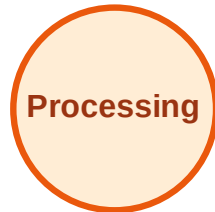
Visited: A, B, D



DFS Traversal

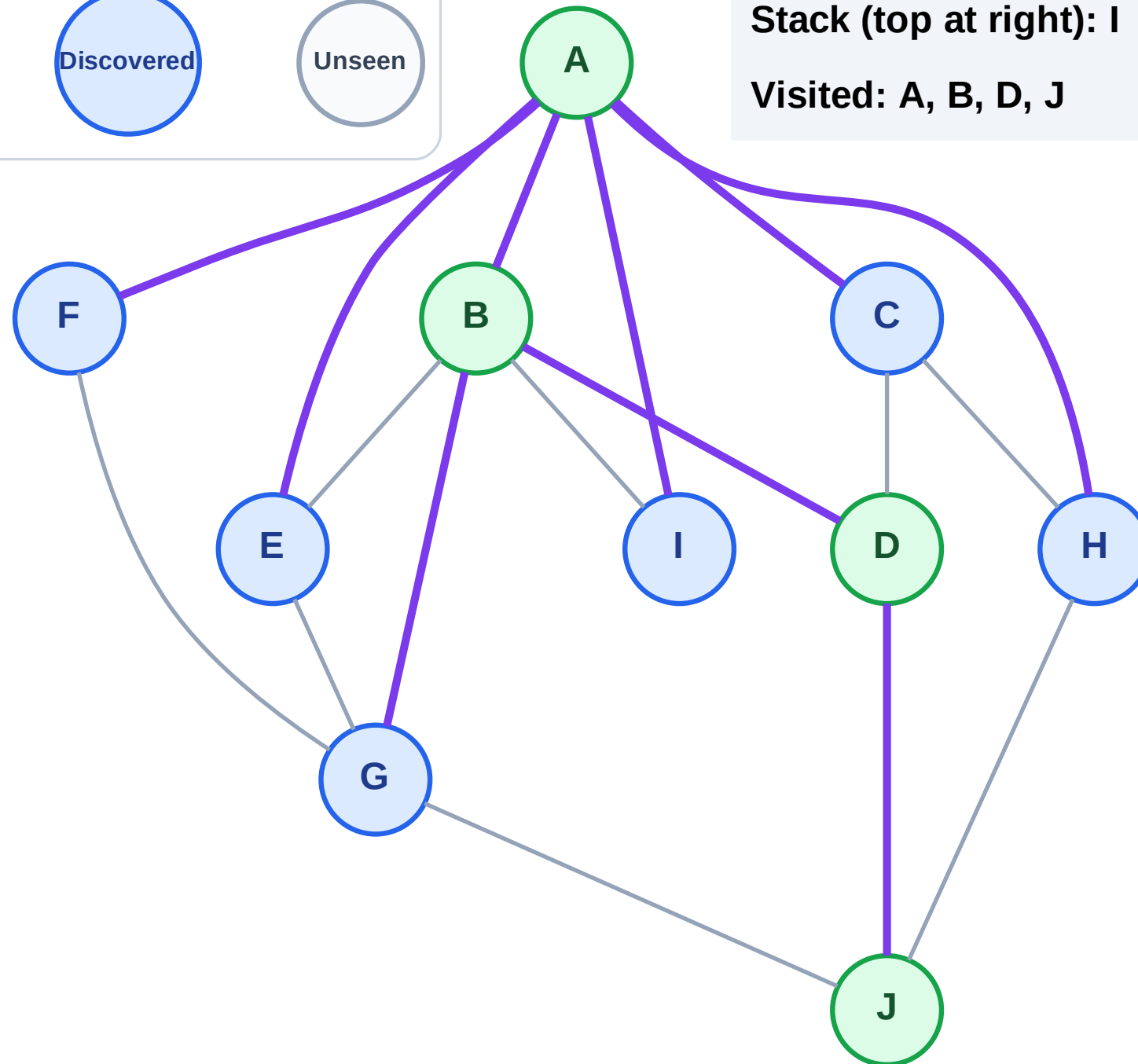
Step 9: No new neighbors from J

Legend



Stack (top at right): I | H | F | E | C | G

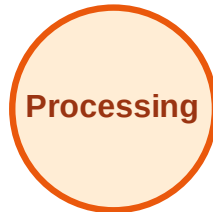
Visited: A, B, D, J



DFS Traversal

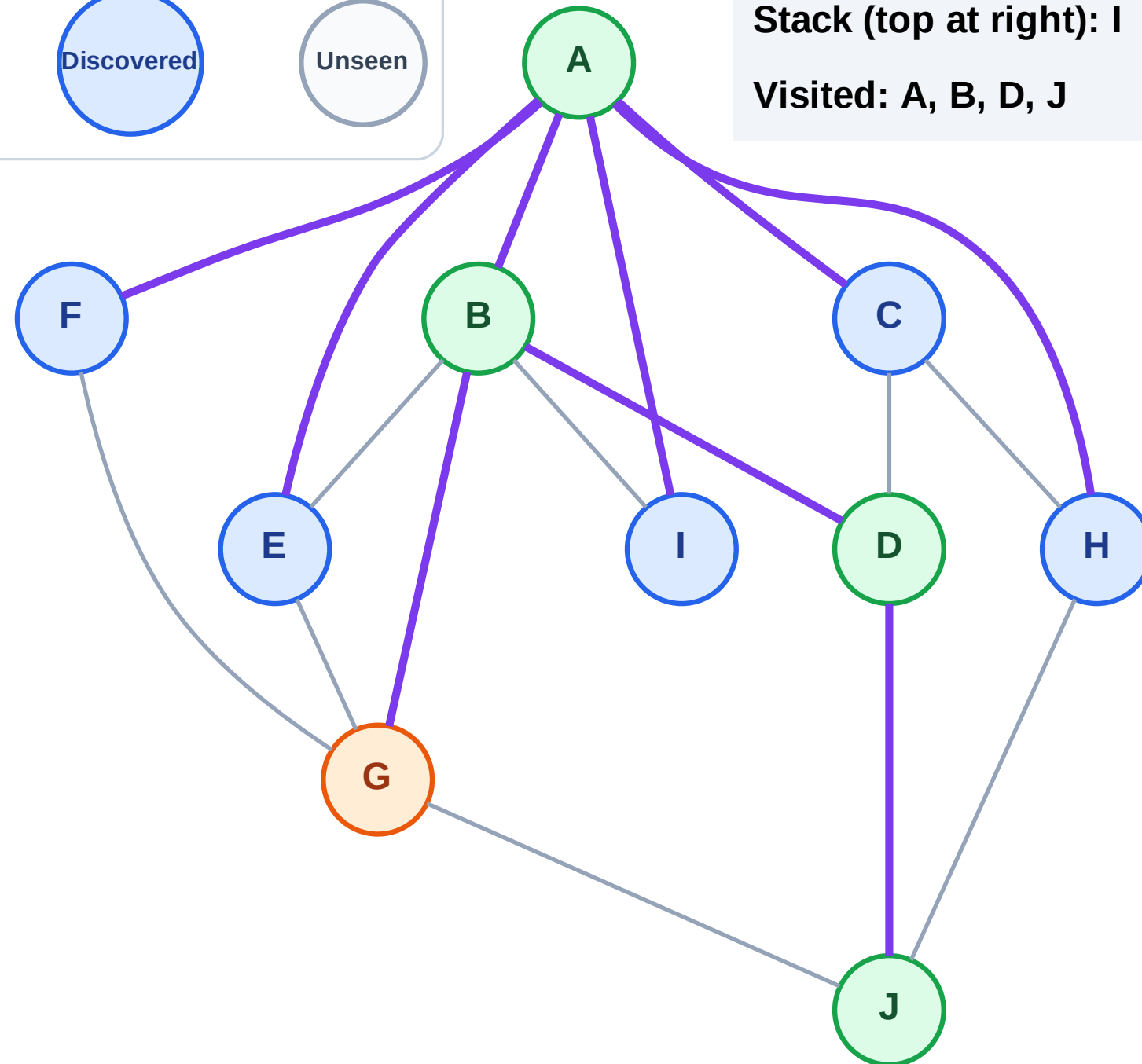
Step 10: Process node G

Legend



Stack (top at right): I | H | F | E | C

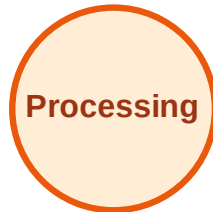
Visited: A, B, D, J



DFS Traversal

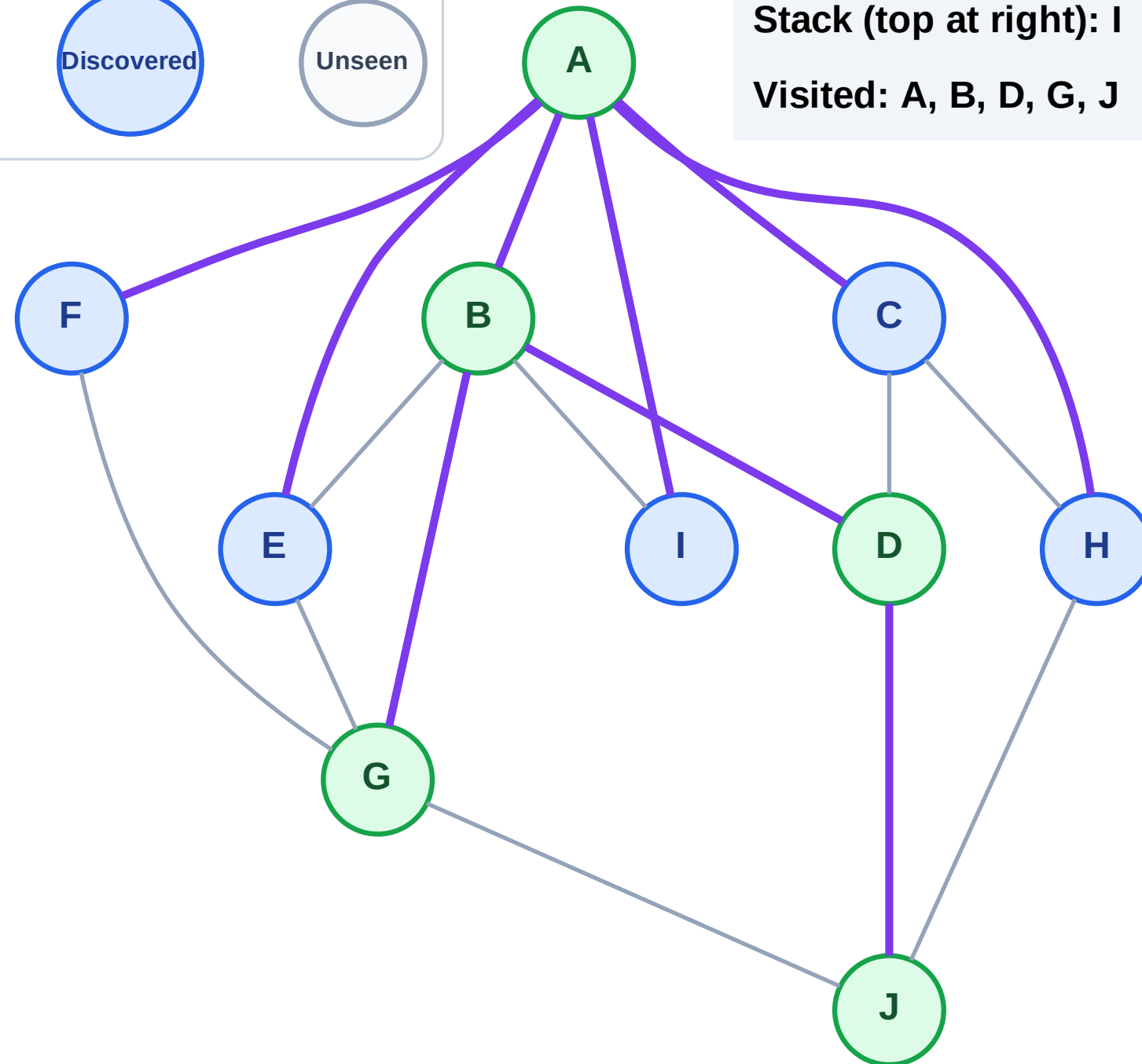
Step 11: No new neighbors from G

Legend



Stack (top at right): I | H | F | E | C

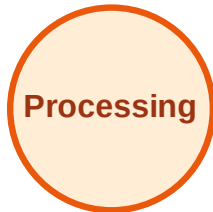
Visited: A, B, D, G, J



DFS Traversal

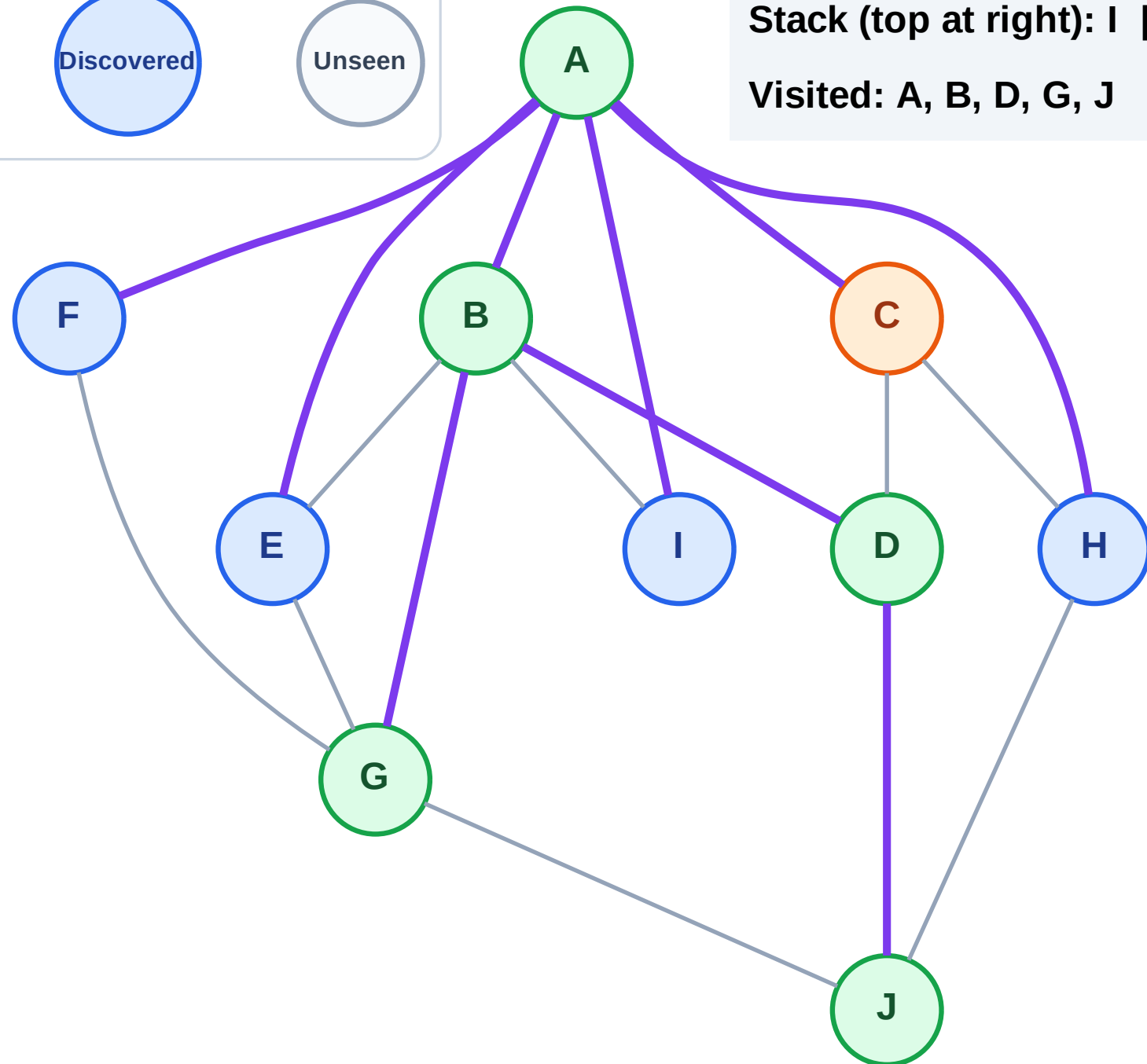
Step 12: Process node C

Legend



Stack (top at right): I | H | F | E

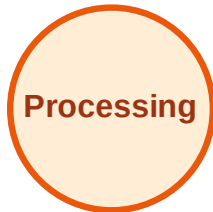
Visited: A, B, D, G, J



DFS Traversal

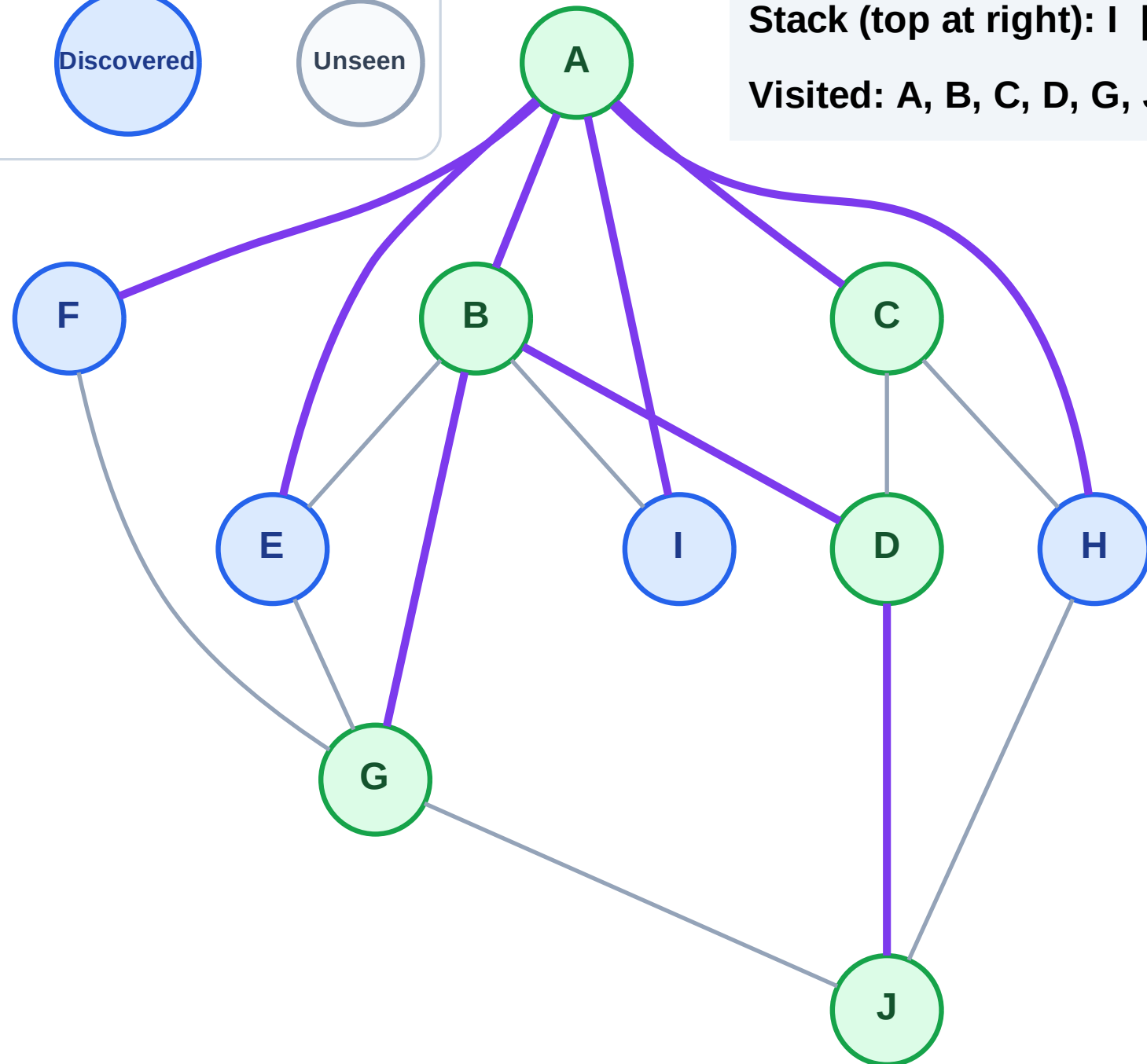
Step 13: No new neighbors from C

Legend



Stack (top at right): I | H | F | E

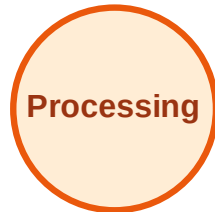
Visited: A, B, C, D, G, J



DFS Traversal

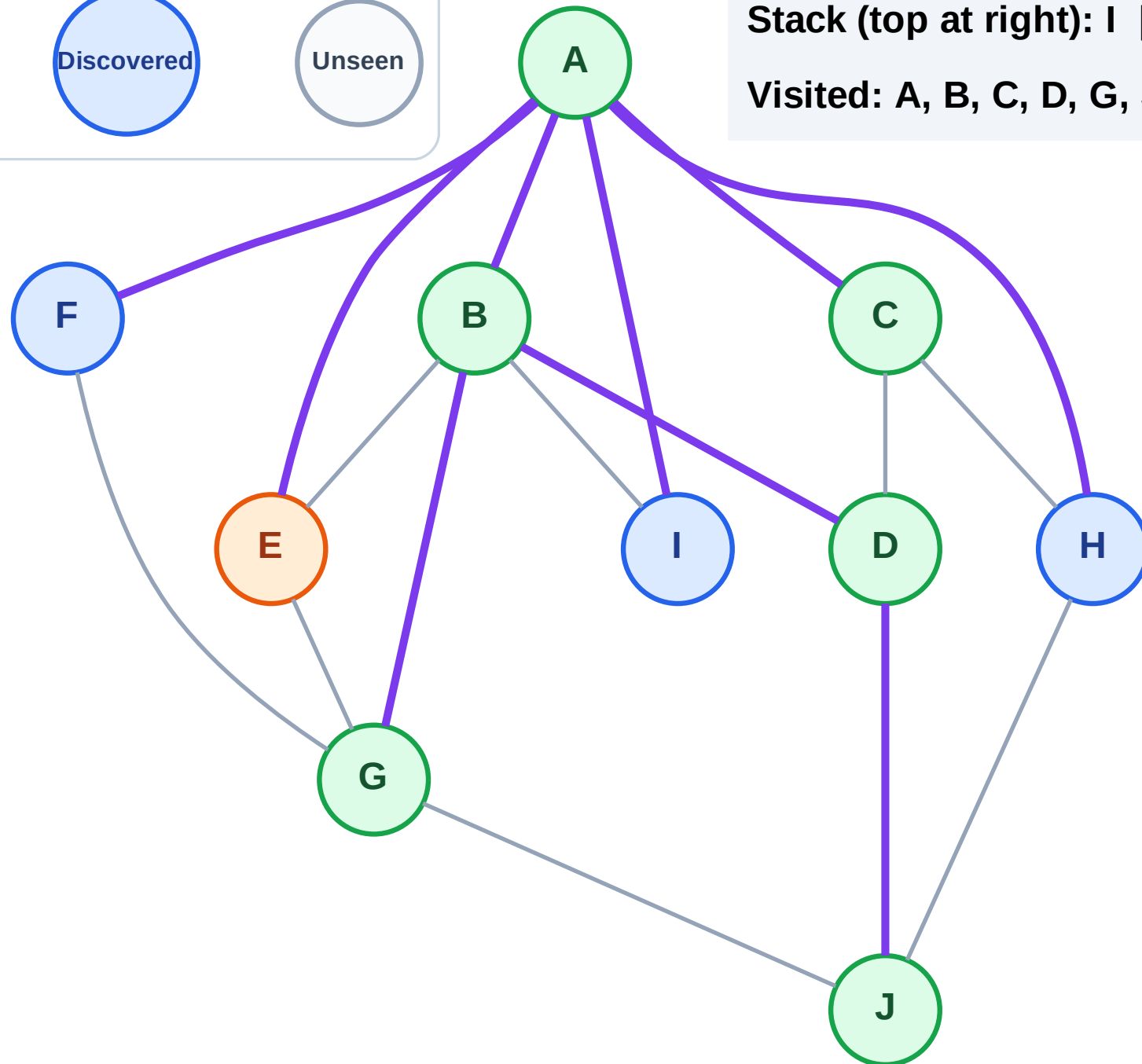
Step 14: Process node E

Legend



Stack (top at right): I | H | F

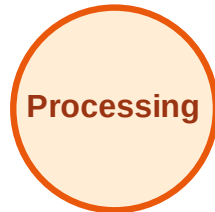
Visited: A, B, C, D, G, J



DFS Traversal

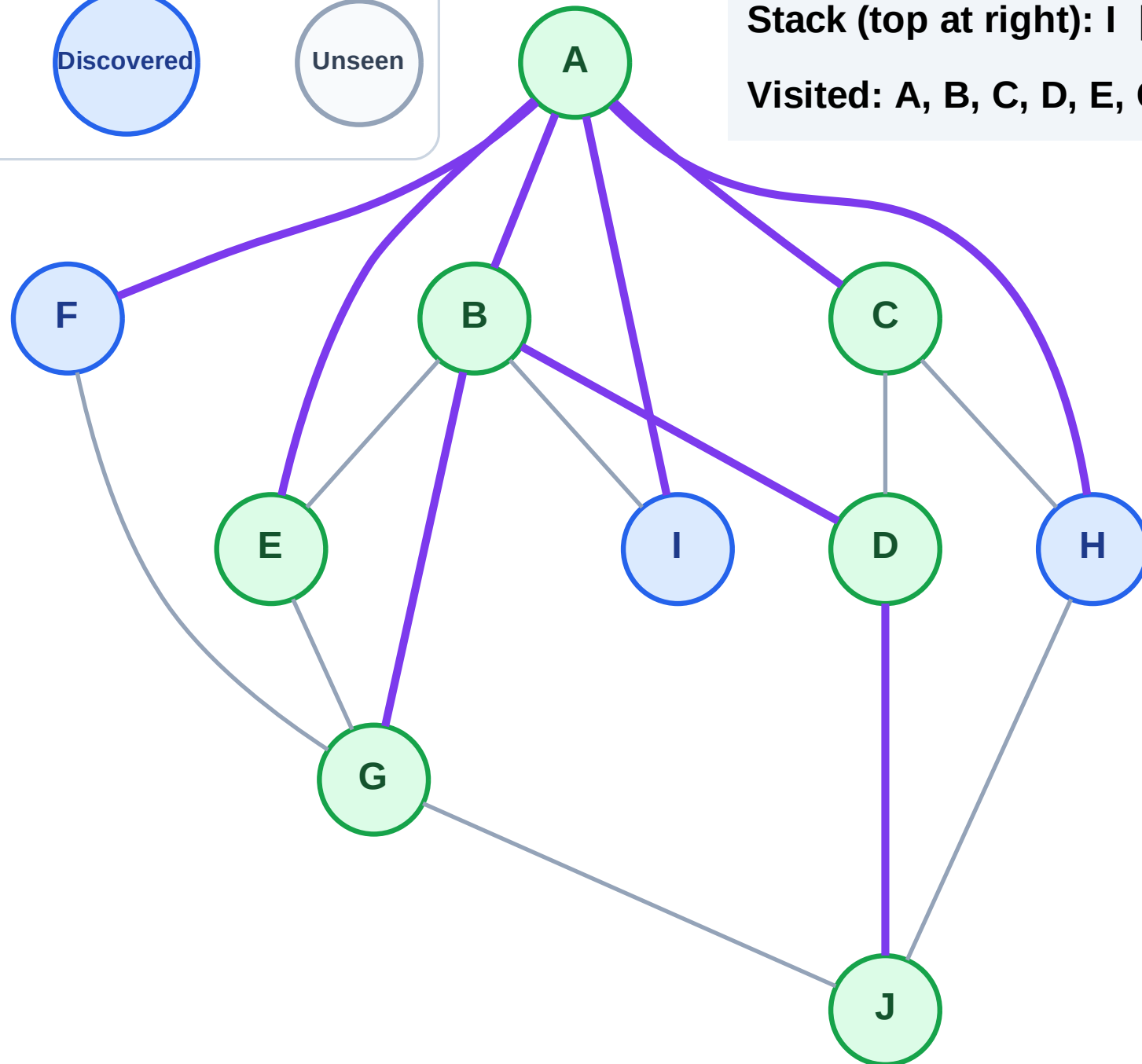
Step 15: No new neighbors from E

Legend



Stack (top at right): I | H | F

Visited: A, B, C, D, E, G, J



DFS Traversal

Step 16: Process node F

Legend

Complete

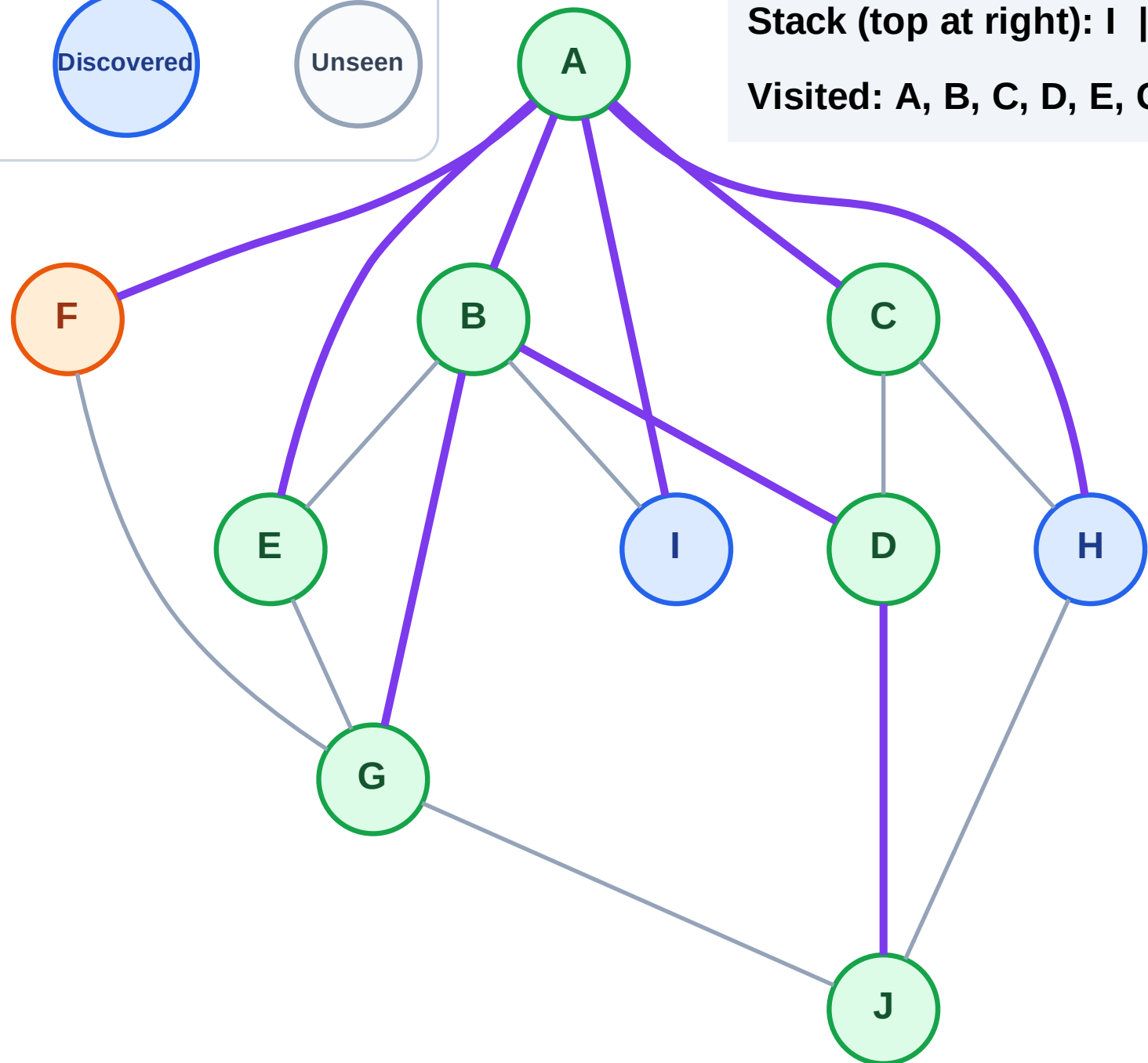
Processing

Discovered

Unseen

Stack (top at right): I | H

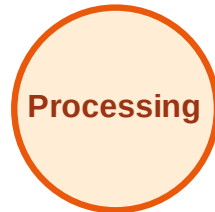
Visited: A, B, C, D, E, G, J



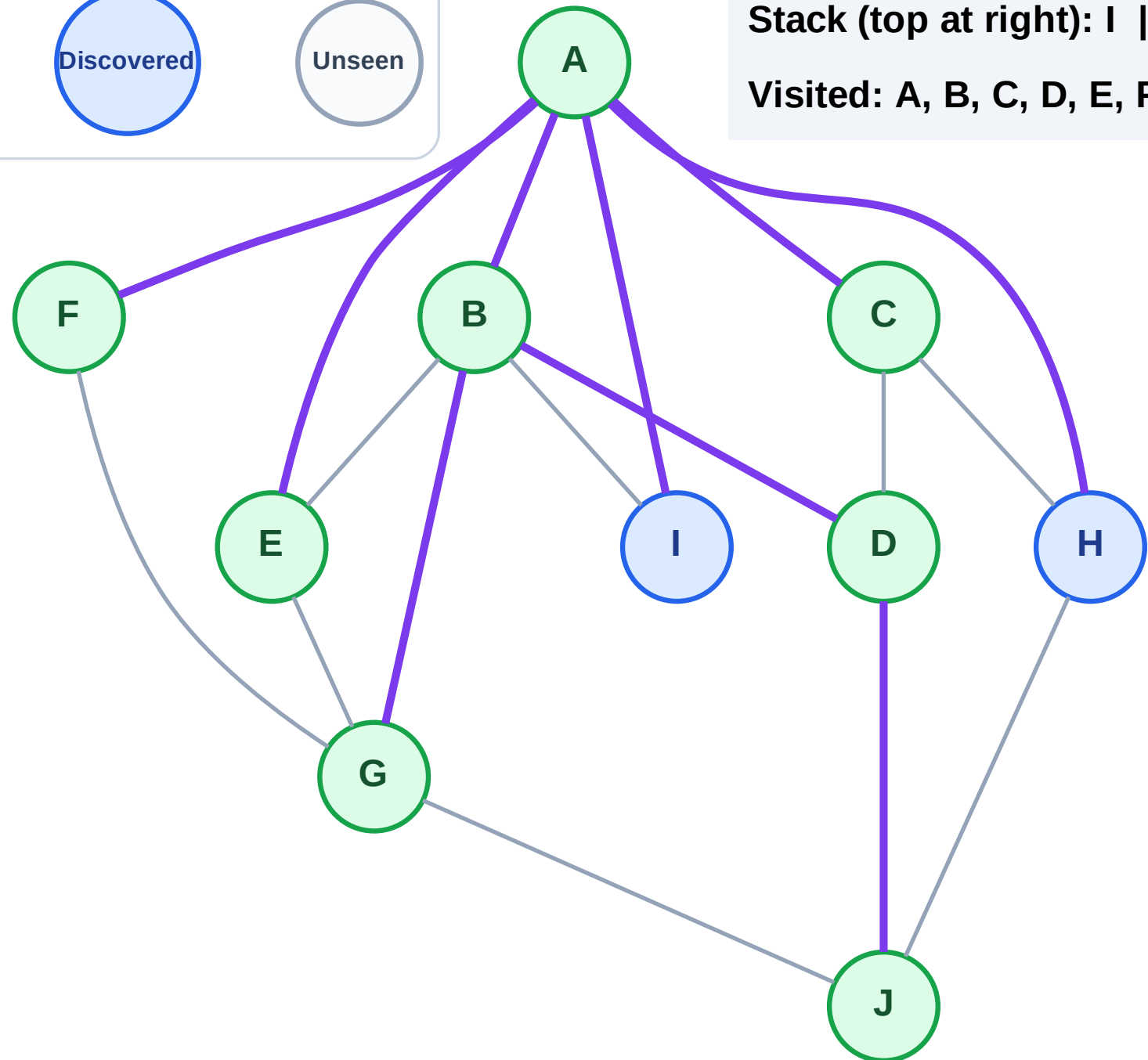
DFS Traversal

Step 17: No new neighbors from F

Legend



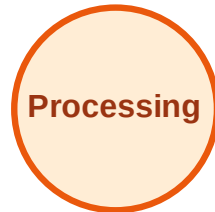
Stack (top at right): I | H
Visited: A, B, C, D, E, F, G, J



DFS Traversal

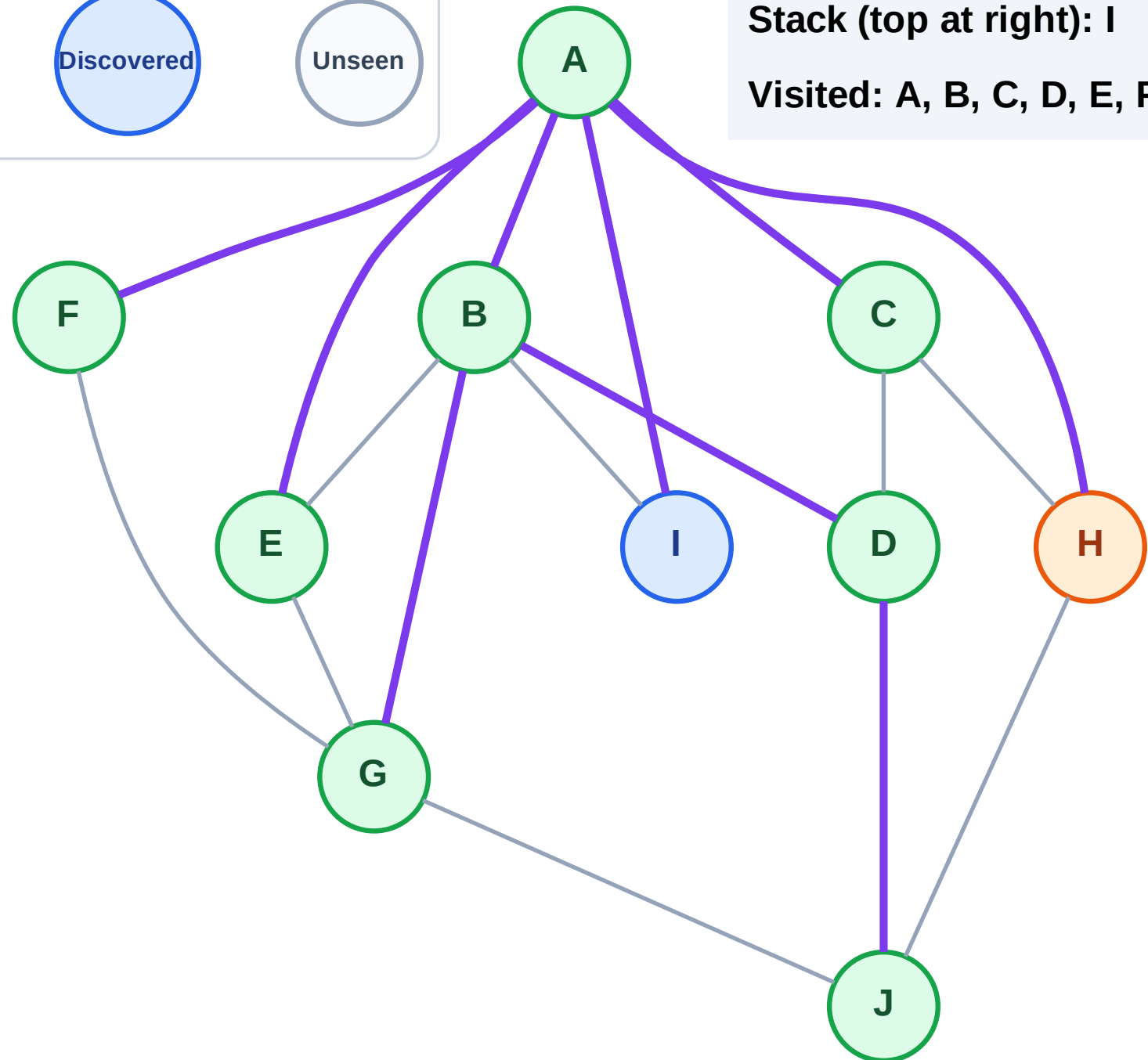
Step 18: Process node H

Legend



Stack (top at right): I

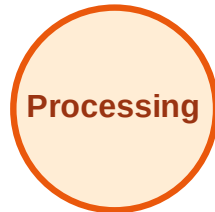
Visited: A, B, C, D, E, F, G, J



DFS Traversal

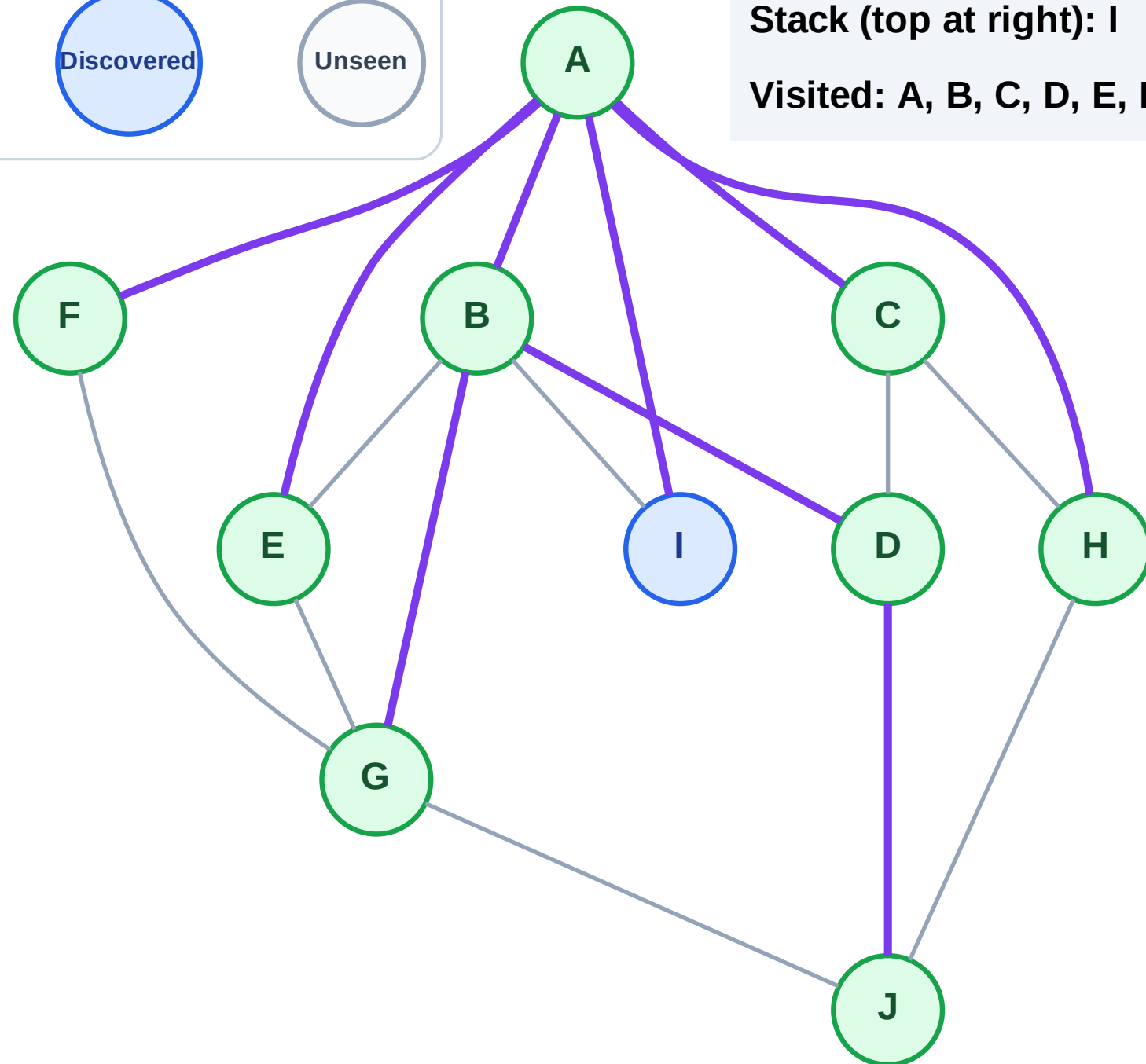
Step 19: No new neighbors from H

Legend



Stack (top at right): I

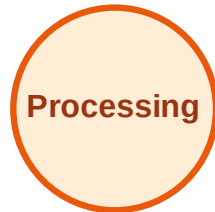
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

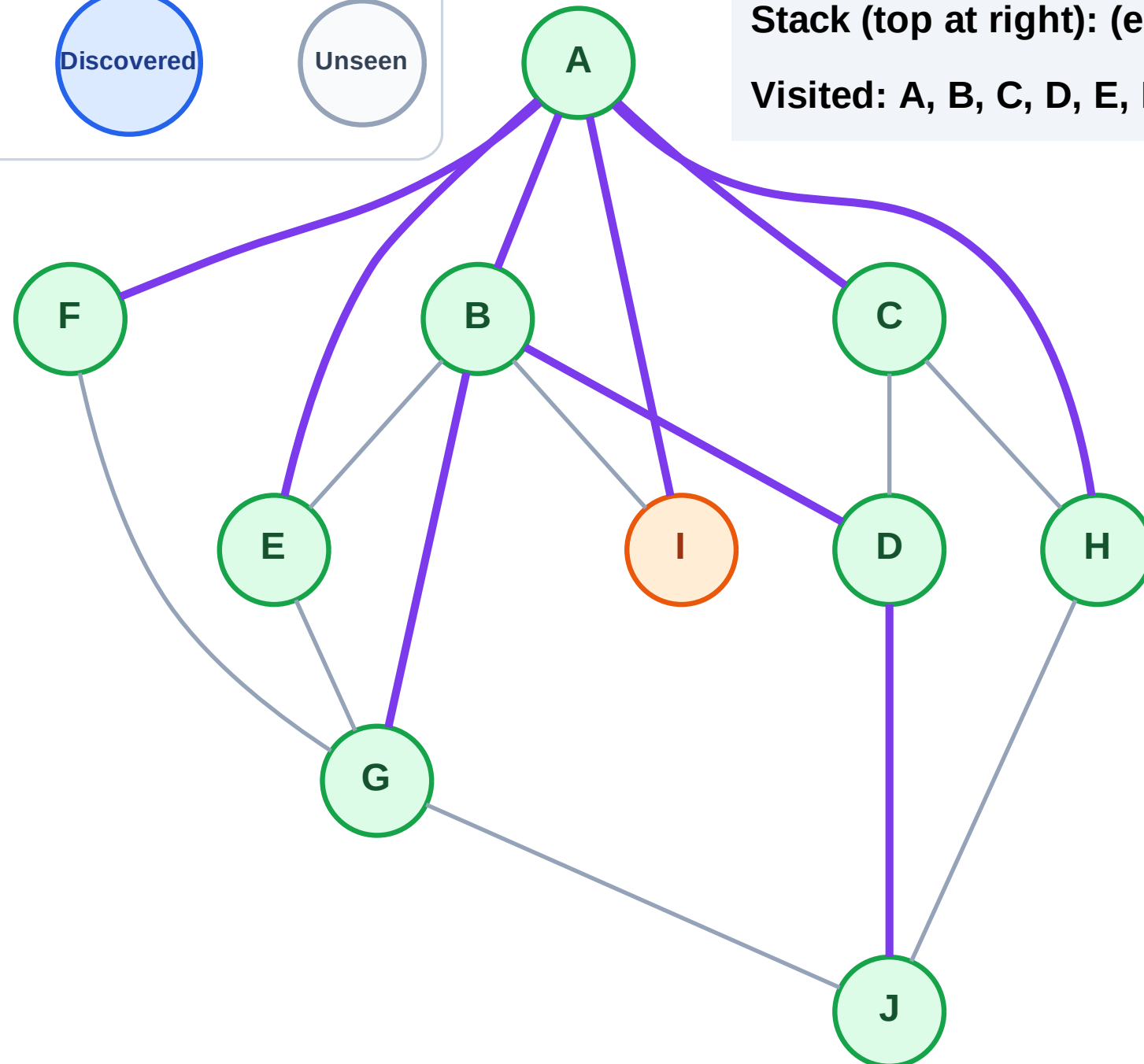
Step 20: Process node I

Legend



Stack (top at right): (empty)

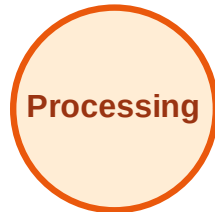
Visited: A, B, C, D, E, F, G, H, J



DFS Traversal

Step 21: No new neighbors from I

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

